

Generative AI for Data Analysis of Physical Structures from Aerial Images and Tabular Data

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Paderborn University



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Documentation

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Official Declaration

I hereby declare that I have authored this thesis independently and without unauthorized assistance. All sources and aids used have been indicated and acknowledged appropriately.

This thesis has not been submitted, either in the same or a similar form, to any other examination authority.

Paderborn, July 25, 2025

A handwritten signature in black ink, appearing to read 'Gunasekhar', with a stylized flourish underneath.

Gunasekhar Jenni

Abstract

This thesis explores the use of Generative AI, particularly Vision Language Models (VLMs), to analyze physical structures from aerial imagery in combination with structured tabular data. The motivation stems from the need to bridge the gap between visual and structured geospatial information, a common challenge in remote sensing and energy related assessments. The study evaluates how well open source VLMs can interpret architectural features such as skylights, dormers, and roof types through zero shot prompting and multimodal reasoning tasks. These visual cues are then linked to corresponding attributes in structured data, enabling insights into use cases like assessing attic livability. The findings reveal that while current models show promise in identifying fine grained features, there are notable limitations in consistency and localization accuracy. This work highlights the potential of multimodal AI to automate and scale geospatial analysis and provides a foundation for future systems that integrate image and tabular data more effectively.

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Introduction

“ *You can’t do better design with a computer, but you can speed up your work enormously.*

— **Wim Crouwel**

(Graphic designer and typographer)

In recent years, the integration of Generative AI into geospatial analysis and energy data processing has created new opportunities to extract actionable insights from complex, multimodal datasets. This is especially relevant when analyzing residential structures through aerial imagery and associated metadata. In such scenarios, data typically consists of tabular records containing detailed information about individual buildings, such as roof type, photovoltaic potential, energy usage, and physical attributes, linked to corresponding aerial images. These images often depict diverse roof types (e.g., gabled, hipped, flat), and the ability to relate visual features to structured data points is essential for accurate evaluation. Traditionally, analysts manually correlate visual inputs with spreadsheet entries, a process that is both inefficient and error prone.

This thesis proposes a novel approach using open source Vision Language Models (VLMs) and Prompt Engineering techniques to perform inference on both image and tabular data. Prompt engineering strategies will guide the model across three types of inference: image only, tabular only, and joint inference using both modalities. The system will link visual characteristics from aerial images, such as roof features or roof type, with specific entries in a dataset, enabling a unified analysis of heterogeneous data sources. To assess the effectiveness of these methods, the outputs generated from each inference type will be systematically compared against predefined ground truth values. This evaluation will help validate the model’s accuracy and performance.

This research demonstrates a practical workflow for harnessing generative AI to fuse visual and structured data in remote sensing applications. It begins by selecting and benchmarking open source vision language models, then shows how to preprocess and annotate high resolution imagery (e.g., marking roof features) alongside extracting complementary attributes from a relational database. By comparing image

only, table only, and combined multimodal inferences, it quantifies the gains that arise when visual features and metadata are jointly leveraged. Although illustrated on residential roof analysis, this end to end pipeline can be readily adapted to other remote sensing use cases, such as land cover classification, infrastructure monitoring, or environmental change detection, where integrating image data with tabular records unlocks richer, more accurate insights.

1.1 Motivation

In energy economics and urban planning, informed decision making depends on analyzing both spatial imagery and structured data, yet these data types are often processed independently. Aerial images offer rich visual cues about buildings, such as shape, orientation, and roof features, while associated tabular datasets contain detailed attributes like roof type, roof slope, roof height, roof area, energy demand, and photovoltaic (PV) potential. Manually aligning this visual and structured information is labor intensive, error prone, and not scalable. The emergence of Generative AI models, specifically open source Vision Language Models (VLMs) and related frameworks, presents a timely opportunity to automate this process. This thesis is motivated by the potential to use prompt engineering and multimodal inference techniques to derive insights from both image and tabular data, individually and jointly, and evaluate these outputs against ground truth values, enabling more accurate and efficient analysis for real world energy applications.

1.2 Problem Statement

Despite the availability of aerial imagery and detailed building metadata, current analytical workflows in the energy and infrastructure sectors lack effective tools to unify these data sources. Analysts are often required to manually associate features observed in images, such as roof type, dormer windows, and skylights, with corresponding entries in spreadsheets containing attributes like roof slope, roof height, and roof area. This manual process is not only time consuming and resource intensive but also introduces a high potential for mismatches and inconsistencies, especially when working with large datasets in urban environments. While recent advancements in Vision Language Models (VLMs) offer the potential to automatically detect visual roof features and classify roof types, these capabilities remain underutilized in domain specific applications such as energy planning. Furthermore,

current AI tools rarely integrate image based inferences with structured tabular data for holistic analysis. There is a pressing need for an automated and explainable system that combines both data modalities to deliver accurate, cross referenced insights. Addressing this challenge would streamline building diagnostics, enhance energy efficiency assessments, and support scalable, data driven decision making in the built environment.

1.3 Objectives of the Research

The primary objective of this research is to develop a robust, multimodal data analysis framework that integrates aerial imagery with corresponding tabular building data using Vision Language Models (VLMs) and Generative AI techniques. The system aims to automate the extraction, interpretation, and alignment of visual and textual information, reducing manual effort and improving the accuracy of building-level insights. In addition to model development, a key goal is to perform thorough data analysis across both modalities to uncover patterns and evaluate the quality and consistency of the inputs.

Specifically, the research seeks to:

- Evaluate the capability of VLMs to identify roof-related visual features such as dormers, skylights, and roof types from aerial images.
- Conduct descriptive and exploratory data analysis on the tabular dataset to understand the distribution, completeness, and correlations of attributes such as roof slope, roof height, and roof area.
- Design and implement prompt-driven inference pipelines that process images and tabular data both independently and in combination.
- Link visual inputs with structured attributes in tabular datasets using shared identifiers .
- Validate the output of model-based inferences against ground truth values to assess performance, and accuracy.

1.4 Research Questions

The research is guided by three overarching questions, each addressing a critical aspect of the proposed multimodal framework. These are further broken down into sub-questions to provide focused direction for analysis and evaluation.

1. How effectively can Vision Language Models (VLMs) extract and interpret features from aerial imagery?

- How accurately can selected open-source VLMs identify and localize fine-grained roof features (dormers and skylights) in aerial imagery of gable and hipped residential roofs in Kassel?
- How do different prompting techniques and system-role formulations affect VLM performance in extracting and interpreting complex roof features for subsequent attic usability classification?
- What are the limitations of current open-source VLMs in processing aerial building imagery?

2. How can visual data be linked with structured tabular data for combined inference, based on Visual Question Answer (VQA) type prompts?

- What methods can be used to associate image-based features with corresponding rows in tabular datasets?
- Does the use of carefully crafted prompts improve the quality, precision, or interpretability of the model's outputs in multimodal inference?
- Does combined inference outperform single-modality analysis in terms of accuracy and relevance?

3. How can the system's outputs be validated against ground truth data to ensure reliability?

- What evaluation metrics are best suited to assess the performance of image-only, table-only, and combined inference?
- How does validation performance vary across different roof types (gable vs. hipped) and individual features (dormer vs. skylight)?
- How do image-only, tabular-only, and combined multimodal validation results compare when assessed against their respective ground truths?

- What are the observed strengths and weaknesses during the evaluation process, and how can the model be improved?

1.5 Thesis Structure

This thesis is organized into the following chapters, each addressing a key component of the research process:

Chapter 2 – Related Work

This chapter reviews the current literature on Vision Language Models (VLMs), prompt engineering techniques, multimodal inference strategies, and remote sensing applications. It identifies gaps in the integration of aerial imagery and tabular data within AI-based frameworks.

Chapter 3 – Methodology

This chapter outlines the overall design and implementation of the proposed approach. It describes the dataset preparation, model selection, prompting strategies, and multimodal inference setup. It also details the use of embedding models and data alignment techniques.

Chapter 4 – Evaluation and Results

This chapter presents the evaluation strategy, including performance metrics and comparison with ground truth values. It includes results from experiments across image-only, table-only, and combined inference tasks using VLMs and prompt variations.

Chapter 5 – Discussion

This chapter interprets the experimental results, highlighting the strengths and weaknesses of the proposed system. It discusses the impact of prompt design, model performance, and the generalizability of findings within the remote sensing and energy domains.

Chapter 6 – Conclusion

The final chapter summarizes the research contributions, outlines limitations, and proposes directions for future work, including potential improvements to prompt strategies, dataset expansion, and integration into operational energy systems.

References and Appendices

The thesis concludes with a comprehensive list of references and appendices containing additional data analysis, implementation details, and supplementary figures or code used in the research.

Related Work

” *A picture is worth a thousand words. An interface is worth a thousand pictures.*

— **Ben Shneiderman**

(Professor for Computer Science)

The integration of visual and tabular data using Generative AI techniques has gained growing attention in recent years, especially with the rise of multimodal models capable of understanding both images and text. However, the combined application of Vision Language Models and prompt engineering to domains such as building analysis, energy diagnostics, and remote sensing remains limited. To position this thesis within the current state of the art, a systematic literature review was conducted covering January 2020 through January 2025 in IEEE Xplore, ACM Digital Library, Semantic Scholar, and Google Scholar. Search terms included “open source Vision Language Models,” “generative AI for remote sensing,” “prompt engineering techniques,” and “geospatial data.” Retrieved records were imported into a reference manager and mirrored in a spreadsheet; duplicates were removed, and titles and abstracts were screened to exclude out-of-scope studies. Full texts of the remaining publications were then reviewed, and the final corpus was organized into four thematic areas: Vision Language Model overview; generative AI frameworks for remote sensing; prompt engineering methods; and aerial image interpretation.

2.1 Open Source Vision Language Models

Vision-Language Models (VLMs) represent a significant frontier in generative artificial intelligence, extending the capabilities of traditional Large Language Models (LLMs) to encompass multimodal inputs, primarily visual data such as images and videos, alongside textual information. [Li+24] The evolution within this domain has progressed from foundational models like OpenAI’s CLIP [Ope21], which pioneered zero-shot visual concept understanding through natural language guidance, to more advanced generative and conversational VLMs, exemplified by systems such

as GPT-4(V) and Gemini[Dee24], capable of complex multimodal reasoning. The rapid proliferation and advancement of open-source contributions have been instrumental in accelerating VLM research. The availability of open-source LLMs, notably LLaMA and Vicuna, has significantly propelled the development of open-source VLMs. This collaborative ecosystem fosters innovation by allowing researchers to build upon established foundational models and freely share novel architectures, datasets, and methodologies. Open-source VLMs offer critical alternatives to proprietary, closed-source systems, thereby democratizing access to cutting-edge AI technologies, enhancing research reproducibility, and facilitating a deeper understanding of their intricate mechanisms.

2.1.1 Architecture

These models are typically built around a multimodal architecture that learns to associate information across these different modalities. Common VLM architectures generally consist of three core components - vision encoder, language encoder (or LLM) and multimodal fusion module.

Vision Encoder

The vision encoder is responsible for extracting features from visual data, transforming raw pixel inputs into structured “visual tokens” or embeddings that can be interpreted by the language model. Most open-source Vision-Language Models (VLMs) implement this component using variants of the Vision Transformer (ViT) architecture, commonly initialized with pre-trained weights from models like CLIP or OpenCLIP.

For instance, LLaVA [Liu25; Zha+25a] employs the CLIP ViT-L/14 backbone, while MiniGPT-4 [Zhu+23] uses Eva-CLIP ViT-G/14. Qwen-VL [Qwe25a; Qwe25b] builds on OpenCLIP’s ViT-bigG, and InternVL2 [Ope25; Int25] integrates its proprietary InternViT encoder. In contrast, Phi-3.5-Vision [Hug25c] utilizes a dedicated image encoder developed in-house.

Beyond standard ViT setups, some models implement advanced strategies such as “anyres,” which involves dividing high-resolution images into smaller tiles to improve the recognition of fine-grained architectural features.

Language Encoder (LLM Backbone)

The language encoder serves as the linguistic core of a Vision-Language Model, responsible for processing and interpreting natural language. In many multimodal systems, the LLM's weights are frozen during initial pre-training to reduce computational demands while maintaining performance. Most VLMs adopt powerful existing LLMs as their backbone. For example, both LLaVA and MiniGPT-4 are built on Vicuna [neo25; gok25], while BLIP-2 incorporates Flan-T5 or OPT. Qwen-VL relies on Qwen-7B [Qwe25a], Phi-3.5-Vision integrates Phi-3 Mini, and InternVL2 employs either internlm2-chat or Qwen2-Instruct models.

Multimodal Fusion Module (Cross-Modal Connection Module)

This component connects the visual and textual modalities by transforming the extracted visual features into a format the language model can process. One common technique involves using linear projection layers to map vision tokens into the LLM's embedding space, as seen in LLaVA and MiniGPT-4 [neo25; Che+23; gok25]. Alternatively, BLIP-2 and MiniGPT-4 also employ a lightweight Transformer known as a Querying Transformer (Q-Former), which generates fixed-size query embeddings to bridge the modalities efficiently [Hug25a; Vis25]. Cross-attention mechanisms are used in models like Qwen-VL and IDEFICS [Hug25b], enabling the LLM to attend directly to visual features. InternVL2, by contrast, employs a simpler Multi-Layer Perceptron (MLP) as a projection module.

Understanding these architectural components is essential to this thesis, as the proposed methodology employs prompt-based zero-shot inference in which an aerial image of a rooftop and a natural language prompt (e.g., “Does this building have usable attic space?”) are jointly processed by the model. The model's capacity to integrate and reason across visual and textual modalities is fundamentally determined by the architectural design discussed in this section. Consequently, this overview serves as a technical foundation for evaluating the extent to which different open-source VLMs are structurally equipped or constrained in performing the image-to-text inference tasks central to aerial structure analysis.

2.1.2 Training Data

Vision-Language Models (VLMs) are typically trained on vast and diverse datasets that combine visual and textual information. The quality, scale, and characteristics

of this training data are crucial for the models' ability to understand and generate content across modalities.

Commonly used training data characteristics and datasets include:

Large-Scale Image-Text Pair Datasets

To enable general-purpose visual understanding, many Vision-Language Models are initially pre-trained on massive collections of image-text pairs harvested from the internet. While large in scale, such datasets tend to be noisy, requiring careful filtering and bootstrapping. Among the most prominent is LAION, particularly the LAION-5B variant [Sch+22], which serves as a foundation for models like MiniGPT-4, Qwen-VL, and BLIP-2. The CLIP model [Vox24] set a precedent by training on 400 million web-sourced pairs.

Other widely used resources include Conceptual Captions datasets-CC3M and CC12M [Ng+20] featured in the training pipelines of MiniGPT-4, LLaVA, VLMo, and GLIP [Mic25]. The SBU Captions dataset [Ord25] is similarly common, as is COCO [Lin+14], which contributes to both pre-training and fine-tuning stages for Qwen-VL, BLIP-2, and VLMo. Visual Genome [Kri+17] enhances fine-grained reasoning and is integrated into Qwen-VL and VLMo, while Flickr30k [Liu+24b] finds usage in GLIP and VLMo.

High-Quality, Reasoning-Dense Data

To overcome the limitations of noisy web data, modern approaches emphasize reasoning-dense, high-quality datasets-often synthetic or curated. For example, LLaVA integrates GPT-generated instruction-following data tailored for multimodal reasoning. MiniGPT-4 takes a hybrid approach, generating refined image-text pairs through iterative collaboration between the model and ChatGPT. Similarly, Phi-3.5-Vision relies on carefully filtered synthetic datasets and public websites curated for logical consistency and reasoning richness.

Multi-Modal and Interleaved Data

Recent VLMs increasingly leverage datasets that combine interleaved images and text, or incorporate multiple modalities beyond just vision and language. IDEFICS exemplifies this trend by training on natural documents with image-text interleaving,

such as those from the OBELISC dataset comprising 141 million web pages, 353 million images, and over 115 billion text tokens. Qwen-VL extends this capability by supporting arbitrary interleaved image-text inputs for multi-image scenarios. InternVL2 goes further by incorporating long documents, multiple images, medical data, and video streams. NExT-GPT [NEx25] advances this multimodality by handling inputs and outputs in various combinations of text, image, video, and audio.

Task-Specific Datasets for Fine-Tuning

After broad pre-training, models are typically fine-tuned on smaller, task-specific datasets that help align their capabilities with downstream use cases. MiniGPT-4, for instance, is fine-tuned on just 3,500 carefully designed conversation pairs to improve its generation stability and contextual alignment. LLaVA-Phi adopts a similar strategy by incorporating instruction-tuning datasets tailored to visual question answering and user-driven interactions.

The nature and scope of the training data used to pretrain vision-language models are critical to understanding their applicability to this research. Most open-source VLMs are trained on large-scale image-text datasets derived from general web content, such as LAION or COCO, which predominantly feature natural scenes, everyday objects, and human-centric imagery. However, these datasets typically lack domain-specific aerial imagery and architectural elements such as dormers, skylights, or varying roof slopes. As a result, the models may struggle to accurately extract fine-grained structural features from rooftop images, which is an essential requirement in this study. Therefore, this section underscores the potential domain gap between general-purpose training data and the specialized requirements of geospatial inference tasks addressed in the research.

2.1.3 Model Parameters

Open-source Vision-Language Models (VLMs) vary widely in parameter size, with some designed for resource-constrained environments and others optimized for comprehensive multimodal understanding. The number of parameters directly influences model performance, hardware requirements, and real-world deployability.

Overview of Parameter Scales in Open Source VLMs

LLaVA provides several variants across different scales. LLaVA-1.5 contains approximately 13 billion parameters, while smaller alternatives such as LLaVA-Phi are limited to around 3 billion. The OneVision series offers even more flexibility, with configurations ranging from 0.5 billion to 72 billion parameters.

Qwen-VL also offers a range of sizes, including 3 billion, 7 billion, and 72 billion parameter models. Its successor, Qwen2-VL, comes in 2 billion and 7 billion parameter variants. The more recent Qwen2.5-VL further broadens this spectrum, offering models with 3, 7, 32, and 72 billion parameters.

Phi-3.5-Vision adopts a lightweight design with a 4.2 billion parameter configuration that combines a dedicated image encoder and the compact Phi-3 Mini language model.

InternVL2 spans an extensive range, from 1 billion to 108 billion parameters. A representative mid-size example is InternVL2-8B, which features an InternViT-300M-448px vision module paired with an internlm2_5-7b-chat language component.

The original CLIP models are comparatively smaller, ranging from 63 million to 355 million parameters. A specific example is the ViT-B/16 checkpoint, where the CLIPTextEncoder contains 76 million parameters and the CLIPImageEncoder includes 115 million.

MiniGPT-4 takes a parameter-efficient approach by aligning a visual encoder with a frozen language model, Vicuna-13B, using only a single trainable projection layer. This design significantly reduces the number of trainable parameters without sacrificing multimodal performance.

Impact of Parameter Count on Performance and Deployment

Larger models generally exhibit better accuracy and more sophisticated capabilities, but they also demand significantly more computational resources for both training and inference, leading to increased test times and reduced inference speeds. This trade-off is particularly relevant for time-sensitive or real-time applications.

The number of model parameters plays a significant role in determining a vision-language model's capacity for multimodal understanding, particularly in zero-shot or prompt-based settings. Larger models generally possess stronger representational capabilities and a greater ability to capture nuanced relationships between image and

text modalities. This is relevant to the present study, as the inference tasks explored, such as detecting dormers, interpreting skylight features, and associating these with tabular roof attributes, require the model to process subtle visual cues and respond meaningfully to domain-specific prompts. However, increased parameter size also implies higher computational requirements, which may constrain deployment and batch inference feasibility in practical applications. Thus, the examination of model scale in this section provides critical context for evaluating trade-offs between inference performance and usability in the context of this research.

2.1.4 Licensing and Accessibility

The licensing landscape for open-source Vision-Language Models (VLMs) is highly diverse, influencing not only their accessibility and modifiability but also their suitability for commercial use. A clear understanding of these licensing terms is essential for researchers and developers intending to adopt or build upon existing models.

One of the most permissive and commonly used licenses is the MIT License. It allows for commercial use, modification, distribution, and private deployment, with the primary condition being the preservation of copyright and license notices. Several well-known models fall under this category. Phi-3.5-Vision, BLIP-2 (specifically the Salesforce/blip2-opt-2.7b variant), and InternVL2 are all released under the MIT License. Similarly, GLIP is also MIT-licensed. While OpenAI's original CLIP implementation uses a more restrictive license, some derived implementations, such as Qualcomm's optimized version, adopt the MIT License for specific components.

Another widely adopted permissive license is Apache 2.0, which additionally includes patent grants. This license is employed by many models in the Qwen2.5-VL series, including the Qwen2.5-VL-32B-Instruct and Qwen2.5-Omni-7B variants. The notable exception is the 72B variant, which may have alternative licensing terms.

Not all models are open for commercial usage. Some VLMs are explicitly restricted to research-only or non-commercial contexts, either due to the license governing the model weights or the underlying datasets. For instance, PandaGPT's dataset and delta weights are licensed under CC BY-NC 4.0 (Creative Commons Attribution–NonCommercial 4.0 International), which restricts usage to non-commercial academic or research scenarios [yxu25]. Likewise, NExT-GPT, despite using a BSD 3-Clause license, is labeled for non-commercial use only, with any commercial application requiring explicit permission from the authors.

The implications of these licensing terms are profound. Permissive licenses like MIT and Apache 2.0 foster widespread adoption and integration into commercial products, accelerating innovation across industries. Conversely, research-only or non-commercial licenses, while promoting academic progress and responsible AI development, limit direct commercial exploitation without specific agreements.

Licensing terms and accessibility constraints are important factors when selecting vision-language models for research and deployment. Many high-performing VLMs are released under non-commercial or research-only licenses, which may limit their applicability in real-world or enterprise-level geospatial applications. Additionally, some models are restricted by API-only access or regional limitations, which can hinder reproducibility, customization, and integration into local inference pipelines. This is particularly relevant to the goals of this thesis, which aims to conduct transparent, scalable evaluations of open models for analyzing aerial imagery and structured building data. The licensing and deployment considerations discussed in this section inform the selection of models that can be used effectively within the constraints of academic research, while also highlighting potential limitations for future applied use cases.

2.1.5 Benchmarking of VLMs

VLMs are rigorously evaluated against a suite of benchmark datasets that provide standardized tasks for performance assessment. These datasets typically contain labeled examples with well-defined expected outputs, enabling quantitative measurement of model performance.

Key benchmarking datasets include [Mil25; IBM25]:

- **Visual Question Answering (VQA):** VQA v2 is a widely used benchmark containing over 1.1 million questions tied to 200,000 real-world images, testing object recognition, spatial relationships, and scene context. Derivatives include GQA (compositional questions requiring logical inference), OK-VQA (requiring external knowledge), ScienceQA (science question answering), and TextVQA (reasoning based on text in images). Models are scored based on answer accuracy.
- **COCO Captions:** This benchmark assesses image captioning quality, with models generating descriptive text for images. Metrics like BLEU, METEOR, and CIDEr compare generated captions to human-written references, emphasizing natural language fluency and alignment with visual content.

- **Cross-modal Retrieval Tasks:** Datasets like Flickr30k and MS-COCO measure how effectively models associate images with relevant text and vice versa. Evaluation often uses recall@k, indicating how often the correct result appears in the top k matches .
- **Massive Multidiscipline Multimodal Understanding (MMMU):** Contains multimodal multiple-choice challenges across various subjects to measure knowledge, perception, and reasoning skills .
- **MMBench:** Features a collection of multiple-choice questions covering various evaluation dimensions, including object localization and optical character recognition (OCR) .
- **MM-Vet:** Assesses the integration of different VLM capabilities, such as language generation and spatial awareness .
- **OCRBench:** Focuses specifically on the OCR abilities of VLMs, comprising components like document-oriented VQA, handwritten mathematical expression recognition, key information extraction, text recognition, and scene text-centric VQA .
- **MathVista:** A benchmark for visual mathematical reasoning .

Benchmarking provides a standardized means of evaluating and comparing the performance of vision-language models across a variety of tasks. However, most existing benchmarks are focused on general-purpose domains such as image captioning, visual question answering, or reasoning over everyday scenes. These tasks often do not reflect the unique demands of geospatial inference, where models must extract fine-grained architectural details from aerial views and reason about structured metadata. This gap is directly relevant to the third research question of this thesis, which evaluates the reliability of model outputs against ground truth conditions.

2.1.6 Vision Tasks

Vision-Language Models (VLMs) are increasingly recognized as powerful tools for a broad spectrum of computer vision tasks, thanks to their ability to understand natural language and interpret visual data. Their multimodal flexibility allows them to process inputs across video, image, and text formats and to generate meaningful, often context-aware, textual outputs [Hug25a; Li+22; Ran+25; Enc25].

One of the most prominent use cases is Visual Question Answering (VQA), in which the model responds to open-ended questions about visual inputs, showcasing its reasoning capabilities. Closely related is image captioning, where VLMs generate detailed textual descriptions for given images, effectively translating visual semantics into natural language. In the task of image-text retrieval, these models associate images with corresponding textual descriptions and vice versa, enabling applications such as semantic search and cross-modal retrieval.

VLMs also perform object detection and localization by identifying and spatially delineating objects within an image, typically through bounding boxes or key points. Visual reasoning extends these abilities by enabling the model to infer relationships between visual elements—such as object interactions or spatial arrangements—and draw logical conclusions based on those observations.

The vision tasks that VLMs are typically trained or evaluated on, such as image captioning, object detection, and visual question answering, provide insight into the types of visual reasoning these models are capable of. However, these tasks often emphasize natural scenes and broad object categories rather than fine-grained architectural analysis from aerial perspectives. This distinction is central to the first research question in this thesis, which investigates how effectively VLMs can identify and interpret rooftop features like dormers and skylights. Understanding the alignment and mismatch between common vision tasks and the requirements of geospatial roof analysis helps contextualize model performance and informs prompt design choices tailored to this domain.

2.2 Generative AI for Remote Sensing

The field of artificial intelligence has witnessed transformative advancements with the emergence of generative AI, particularly large language models (LLMs) and their multimodal extensions, Vision-Language Models (VLMs). These models represent a paradigm shift from traditional discriminative approaches, which typically classify or detect predefined patterns, to generative frameworks that align language with visual information, enabling more complex and human-like understanding and interaction. This capability holds immense potential for remote sensing (RS), a domain rich in visual data from satellites and aerial platforms, crucial for applications ranging from disaster monitoring and urban planning to agricultural management and environmental observation.

The landscape of remote sensing has been significantly reshaped by the recent emergence of Vision-Language Models (VLMs) and Multimodal Large Language Models (MLLMs), representing a new wave of advancements in artificial intelligence for Earth observation. Models such as GeoChat [Kuc+24], GeoGPT [Zha+24e], RemoteCLIP [Liu+24a], RSGPT [Hu+25], SkyEyeGPT [ZXY25], EarthGPT [Zha+24c], LHRS-Bot-Nova [Li+25], RS-LLaVA [Baz+24], SkySenseGPT [Luo+24], TEOChat [Irv+24], EarthMarker [Zha+24d], GeoLLaVA [Elg+24], RS-MoE [Lin+24], GeoGround [Zho+24], RingMoGPT [Wan+24], GeoPix [Ou+25], GeoPixel [Sha+25], and VHM [Pan+25] are at the forefront of this trend, demonstrating promising performance and enormous potential by aligning language with visual information to enable more complex and human-like understanding of geospatial data. These recent outcomes are pushing the boundaries of what is possible in remote sensing image analysis, facilitating tasks from detailed object recognition and visual question answering to temporal change detection and pixel-level understanding.

2.2.1 Shared Research Objectives

A fundamental commonality across these studies is the overarching goal of bridging the semantic gap between remote sensing imagery and human language. Researchers consistently aim to enable more intuitive and flexible interaction with complex geospatial information. This manifests in several shared objectives:

- **Enabling Natural Language Interaction:** Nearly all models strive to allow users to query, describe, and analyze RS data using natural language prompts, moving beyond traditional, rigid interfaces. This is evident in tasks like Visual Question Answering (VQA), image captioning, and visual grounding.
- **Developing Unified Models for Multi-Task Learning:** A significant trend is the development of generalist VLMs and MLLMs capable of performing a wide array of RS tasks within a single framework. This unification reduces the need for task-specific models, enhancing efficiency and applicability.
- **Addressing Data Scarcity and Quality Issues:** The lack of large-scale, high-quality, and domain-specific image-text datasets for RS is a recurring challenge. Many papers prioritize the creation of new datasets or innovative data generation pipelines to overcome this limitation.
- **Improving Robustness and Generalization:** A consistent objective is to develop models that perform reliably across diverse RS image conditions,

sensor types, and geographic regions, and can generalize to unseen data or tasks.

The shared objectives across recent generative AI research in remote sensing align closely with the motivations behind this thesis. In particular, the emphasis on enabling natural language interaction with remote sensing data mirrors this study's use of prompt-based vision-language inference to assess rooftop usability. The drive toward unified, multitask models reflects the aim of using a single VLM framework to handle diverse tasks such as visual feature extraction and metadata interpretation. Challenges related to domain-specific data scarcity are directly applicable to the thesis context, where high-quality, labeled aerial roof imagery remains limited. These common objectives underscore the relevance of generative multimodal models for addressing the core research questions in this thesis, especially those concerning cross-modal reasoning and generalization to complex, real-world imagery.

2.2.2 Common Methodological Approaches

The pursuit of these shared objectives has led to several common methodological patterns:

- **Leveraging Pre-trained Large Models:** A prevalent strategy [Kuc+24; ZXY25; Hu+25] involves adapting powerful pre-trained LLMs or VLMs from the natural image domain (e.g., LLaVA, Vicuna, InstructBLIP, CLIP) to the RS domain. This capitalizes on their extensive general knowledge and emergent abilities.
- **Instruction Tuning and Supervised Fine-Tuning:** Most models [Hu+25; ZXY25; Zha+24c; Li+25] undergo instruction tuning or supervised fine-tuning on RS-specific datasets. This process aligns the general models with the unique characteristics and tasks of remote sensing imagery, enabling them to follow domain-specific instructions.
- **Efficient Fine-Tuning Techniques:** To manage the computational demands of large models, researchers frequently employ parameter-efficient fine-tuning (PEFT) methods such as Low-Rank Adaptation (LoRA) and Quantized LoRA (QLoRA), or Mixture of Experts (MoE) architectures [Li+25; Elg+24; Lin+24]. These techniques allow for significant performance gains with fewer trainable parameters and reduced training time.
- **Development of Large-Scale, Domain-Specific Datasets:** As highlighted in the objectives, many papers [Kuc+24; Liu+24a; Hu+25; ZXY25; Zha+24c]

introduce new datasets specifically curated for RS VLMs. These datasets often feature human-annotated rich captions, instruction-following pairs, or semi-automated generation pipelines to create large volumes of high-quality, domain-relevant data.

The methodological strategies adopted by recent RS-VLMs inform the technical design of this thesis, particularly the use of pre-trained VLMs adapted through prompt engineering for domain-specific aerial imagery tasks. Techniques such as instruction tuning and parameter-efficient fine-tuning are not directly implemented in this work but contextualize the limitations of zero-shot approaches and point toward future directions for improving performance. The creation of large-scale, domain-specific datasets also supports the need for more representative training and evaluation sets tailored to roof structures, which is crucial for validating the multimodal model's reliability as explored in the third research question.

2.2.3 Key Differentiating Factors

While commonalities provide a foundation, the distinct contributions of each model often lie in their specialized focus and innovative solutions to particular RS challenges:

- **Task Granularity:**
 - *Image-level understanding:* Models like RSGPT [Hu+25] and RS-LLaVA [Baz+24] primarily focus on image captioning and VQA, providing broad scene descriptions or answers to image-level questions.
 - *Region/Object-level understanding:* GeoChat [Kuc+24], EarthMarker [Zha+24d], and GeoGround [Zho+24] excel in visual grounding, localizing specific objects or regions using bounding boxes or referring expressions.
 - *Pixel-level understanding:* GeoPix [Ou+25] and GeoPixel [Sha+25] push the boundaries to pixel-level segmentation, generating precise masks for objects, which is crucial for fine-grained analysis.
- **Modality Handling:**
 - Most models primarily deal with optical remote sensing images. However, EarthGPT [Zha+24c] stands out for its ability to comprehend multi-sensor RS imagery, including optical, Synthetic Aperture Radar (SAR), and infrared data, enhancing its versatility in diverse conditions.

- **Temporal Reasoning:**

- A significant differentiation is the capability to handle temporal sequences. TEOChat [Irv+24] and GeoLLaVA [Elg+24] are specifically designed for temporal Earth observation data, enabling tasks like change detection and damage assessment by processing image pairs or video frames over time. RingMoGPT [Wan+24] also incorporates a change detection module for multi-temporal analysis.

- **Addressing Specific Model Limitations:**

- *Hallucination and Honesty:* VHM [Pan+25] directly tackles the issue of model hallucination by introducing an “honest” instruction dataset (HnstD) that includes deceptive questions, ensuring the model provides truthful answers even to nonsensical queries. RS-LLaVA [Baz+24] also aims to mitigate hallucination through model collaboration.
- *Multi-scale Objects and High Resolution:* GeoPix [Ou+25] integrates a Class-wise Learnable Memory (CLM) module to facilitate segmentation of multi-scale objects. GeoPixel [Sha+25] specifically supports high-resolution imagery up to 4K HD through an adaptive image divider, crucial for detailed RS analysis.
- *Computational Efficiency:* RS-MoE [Lin+24] leverages a Mixture of Experts architecture with an Instruction Router and lightweight LLMs to enhance efficiency and scalability for caption generation and VQA.
- *Data Generation Philosophy:* Approaches vary from purely human-annotated, rich-content datasets (RSICap in RSGPT [Hu+25], VersaD in VHM [Pan+25]) to semi-automated pipelines leveraging existing data and LMMs (GeoPixelD) or heuristic conversions (RemoteCLIP). The quality and richness of these datasets directly influence the model’s capabilities.

The differentiating factors among RS-VLMs highlight the diversity of design choices and specializations that impact model performance across tasks and modalities. These distinctions provide valuable insight into selecting or evaluating models for this thesis, particularly in terms of task granularity (e.g., image-level vs. pixel-level reasoning), modality support (e.g., optical vs. multisensor), and temporal reasoning capabilities. While this thesis focuses primarily on single-image, optical data, the advances in high-resolution and multi-scale feature segmentation by models like GeoPix and GeoPixel are directly relevant to the goal of fine-grained rooftop feature detection. Likewise, efforts to reduce hallucinations and improve interpretability

contribute to evaluating the model's trustworthiness-an essential aspect of the third research question focused on output reliability.

2.3 Prompt Engineering in Multimodal AI

Prompt engineering has emerged as a pivotal discipline in the era of generative artificial intelligence, focusing on the development and optimization of natural language instructions, or "prompts," to guide AI models toward desired outputs. [Pro25] This discipline encompasses a broad spectrum of techniques, from crafting precise queries and specifying stylistic elements to selecting appropriate vocabulary, providing relevant context, and even defining a character for the AI to emulate. [Wik25c]

For Vision-Language Models (VLMs) and Multimodal Large Language Models (MLLMs), prompt engineering extends this foundational concept by integrating visual inputs alongside textual instructions. This involves structuring inputs in a manner that enables these models to process and reason across mixed textual and visual tokens, thereby bridging the divide toward more intuitive and comprehensive human-computer interaction [Dat25]. The overarching objective is to leverage the formidable capabilities of large pre-trained models, particularly their capacity for zero-shot or few-shot learning, without necessitating extensive fine-tuning or the acquisition of task-specific training data. This approach is especially critical in specialized domains where labeled datasets are inherently scarce, costly, or time-consuming to create.

2.3.1 Definition and Scope of Prompt Engineering

Prompt engineering, in the context of VLMs and MLLMs, is the strategic design of inputs that combine both linguistic and visual modalities to elicit specific, high-quality responses from generative AI models. This discipline is essential for maximizing the utility of models trained on vast datasets of image-text pairs, enabling them to adapt remarkably to various downstream tasks and generalize effectively to novel scenarios. [DSS24] The scope of prompt engineering here is broad, encompassing not only the textual components of a prompt but also the integration and formatting of visual cues [Zha+24a].

The process involves guiding models to perform complex tasks such as question answering, image classification, or even generating new content, by providing

context and instructions through a carefully constructed prompt. This method allows for the efficient use of pre-trained models, reducing the need for extensive task-specific training data, which is a significant advantage in resource-constrained environments. The ability to steer these models through prompts is fundamental to understanding their underlying capabilities and inherent limitations [Siv+24].

The definition and scope of prompt engineering outlined in this section are directly relevant to the methodology employed in this thesis. The ability to strategically structure multimodal prompts, combining aerial imagery with textual instructions, forms the core of the proposed inference approach. Since the thesis investigates zero-shot analysis of rooftop features and their alignment with tabular metadata, the design of prompts becomes essential for guiding the VLM to interpret relevant architectural elements and respond accurately. Understanding the mechanics of prompt construction enables more effective use of general-purpose models for specialized geospatial tasks, which supports both the first and second research questions.

2.3.2 Comparison of Prompt Engineering in Text-Only LLMs vs. VLMs

Prompt engineering strategies differ significantly between text-only Large Language Models (LLMs) and Vision-Language Models (VLMs) or Multimodal LLMs (MLLMs), owing to their varying input modalities, architectural complexities, and interpretive challenges.

In text-only LLMs, prompts are purely textual and primarily involve manipulating linguistic elements such as phrasing, grammar, contextual framing, and reasoning chains (e.g., chain-of-thought) to guide the model toward the desired response [Pro25]. In contrast, VLMs and MLLMs operate on both visual and textual inputs. Prompts in these models are not confined to text but often include visual data such as raw images, bounding boxes, or segmentation masks, enabling richer and more complex instructions [Hon+24; Wik25c].

The constituent elements of prompts also differ. For text-only LLMs, prompt components include queries, instructions, stylistic cues, and context. In VLMs, these are complemented by visual components like images, spatial cues (e.g., bounding boxes, points), and modality-specific attributes. For image generation tasks, prompts may define desired characteristics such as artistic style, lighting, texture, or composition. Word order in such prompts can affect emphasis, and techniques like "negative

prompts" are used to explicitly exclude certain features from the generated output [Wik25c].

Each modality presents unique challenges. Text-only LLMs typically face issues such as ambiguity, vague phrasing, or factual errors, which are primarily linguistic in nature. VLMs, however, must contend with the added complexity of cross-modal alignment. Early VLMs often show limited understanding of negation and complex syntax, requiring careful prompt design and the use of compensatory techniques like negative prompting. The central challenge lies in bridging the semantic gap between textual and visual inputs to avoid misinterpretation or hallucinated outputs [Son+25; Wik25c].

From a technical perspective, text-only LLMs rely solely on linguistic processing, whereas VLMs require advanced mechanisms to fuse visual and textual tokens into a shared embedding space. This is typically achieved through a combination of vision encoders and projectors that convert visual features into a format compatible with the LLM backbone [Zha+24b]. Some architectures go further by using parallel modality learners to prevent attention collapse or shift, ensuring that both modalities contribute meaningfully to the model's reasoning [Hon+24].

Overall, prompt engineering in VLMs is substantially more complex than in text-only LLMs. It is an interdisciplinary task that requires not only crafting well-formed inputs but also an understanding of how multimodal components interact within the architecture. The quality of inference depends directly on how effectively the model can align and process visual and textual information in tandem [Zha+24b; Son+25].

The comparison between text-only and multimodal prompt engineering helps contextualize the added complexity involved in the vision-language setup used in this thesis. Unlike purely linguistic prompts, the prompts in this study combine rooftop images with structured, attribute-oriented questions, requiring careful alignment of visual and textual components. This distinction is important for understanding the challenges and limitations faced during inference, such as misalignment between modalities or hallucinated outputs. The section's insights are therefore essential for refining prompt formats and improving model interaction in the multimodal inference tasks central to this thesis.

2.3.3 Common Prompt Formats

- **Image-Caption Pairs:** This foundational format associates an image directly with descriptive text, often used in pre-training Vision-Language Models (VLMs) to establish correlations between visual content and linguistic representations. For tasks like zero-shot image classification, predefined templates such as "*a photo of a*" are utilized [Zha+24a].
- **Image-Question Prompts:** Predominantly used in Visual Question Answering (VQA) tasks, this format presents an image alongside a natural language question, expecting the VLM to generate a textual answer based on its comprehension of both modalities [Zha+25b].
- **Chain-of-Thought (CoT) Prompting:** Adapted from text-only Large Language Models (LLMs), CoT guides VLMs through intermediate reasoning steps, facilitating complex multimodal reasoning for tasks like conditional referring expressions or automatic prompt optimization [Siv+24].

The overview of common prompt formats provides practical guidance for structuring inputs in this thesis. Specifically, image-question prompts are directly aligned with the core experimental setup, where rooftop images are paired with yes/no or attribute-based queries to infer attic usability.

2.4 Geospatial Data

2.4.1 Definition and Characteristics

Geospatial data refers to information that is associated with a specific location on the Earth. It can be broken down into ‘spatial data’ – data that takes up space – and ‘geo’ – referenced to the Earth. Geospatial data (sometimes just referenced as ‘spatial data’) is used to map and analyze various features and phenomena in the world, providing insights into patterns, relationships, and trends across different spatial environments. This broad category of information is typically classified into two primary types: vector data and raster data (see Figure 2.1).

Vector Data

Vector data utilizes geometric primitives—*points*, *lines*, and *polygons*—to define discrete features with precise spatial locations [Ter25; Pos25].

- **Points** can denote specific locations, such as individual trees or GPS coordinates.
- **Lines** represent linear features like roads or rivers.
- **Polygons** delineate areas, encompassing entities such as building footprints, land parcels, or administrative boundaries.

These features are defined by coordinates, typically in X and Y (or latitude and longitude). The Open Geospatial Consortium (OGC) Simple Features Access standard provides a foundational model for this type of geospatial data, defining geometric types including `Point`, `LineString`, and `Polygon`, along with their multi-part counterparts like `MultiPoint` and `MultiPolygon`.

Coordinates within these geometries can incorporate optional Z -ordinates to represent elevation and M -ordinates to convey measure values, such as time or distance, thereby enabling comprehensive 3D and 4D representations. Each geometric entity is intrinsically linked to a **Spatial Reference System (SRID)** that defines its coordinate system, whether it is a planar system (e.g., easting and northing) or a geodetic system (e.g., longitude and latitude).

Raster Data

In contrast, raster data consists of a continuous grid of pixels, where each pixel holds a value representing a specific attribute. This format is particularly well-suited for continuous phenomena, including elevation, temperature, or land cover. Common examples include satellite images, weather radar maps, and Digital Elevation Models (DEMs).

Additional Considerations

Beyond these fundamental types, geospatial data encompasses a wide array of information, including street networks, climate data, population densities, and land use classifications. The integration of this diverse data facilitates sophisticated

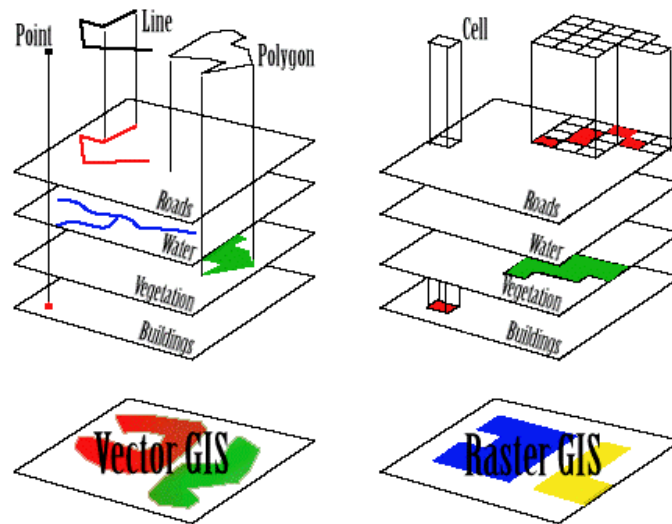


Fig. 2.1: Comparison of Vector GIS versus Raster GIS data models.

spatial analysis, visualization, and informed decision-making by revealing inherent geographic relationships and patterns.

2.4.2 Representation of Spatial Properties of Physical Structures

In the context of urban infrastructure, geospatial data is primarily utilized to construct 3D city models. These digital representations of urban areas encompass terrain, sites, buildings, vegetation, infrastructure, and related objects, all rendered in three-dimensional scale. Such models are described using both two-dimensional (2D) and three-dimensional (3D) spatial data, along with geo-referenced information, making them invaluable tools for presentation, exploration, analysis, and management across diverse application domains. Their particular strength lies in their ability to visually integrate heterogeneous geoinformation within a unified framework, thereby creating and managing complex urban information spaces.

A cornerstone for representing the spatial properties of physical structures is **CityGML**, an Open Geospatial Consortium (OGC) standard. CityGML is an explicit XML-based exchange format that supports not only detailed geometric descriptions but also allows for the specification of rich semantics and topology information for 3D city model components. This spatio-semantic model is critical for enabling complex analyses, as it can reflect the level of building parameter availability through its **Levels of Detail (LoD)**.

Levels of Detail (LoD) are a crucial concept within CityGML, allowing 3D city models to be constructed and represented at various resolutions and abstraction levels. CityGML defines five standard LoDs for building models (see Figure 2.2), ranging from simple 2.5D footprints (LoD 0) to highly detailed architectural models with interior features (LoD 4). This hierarchical approach facilitates efficient visualization and data analysis by providing appropriate levels of complexity for different applications.

- **LoD 0** represents 2.5D building footprints, typically derived from cadastral data.
- **LoD 1** depicts buildings as block models, usually extruded footprints, providing a basic 3D form.
- **LoD 2** building models include standard roof structures, offering a more detailed representation suitable for analyses like solar potential mapping and energy demand calculations.
- **LoD 3** features detailed architectural building models, incorporating intricate wall and roof structures, balconies, and bays.
- **LoD 4** extends LoD 3 models by adding interior features such as rooms, internal doors, and furniture.



Fig. 2.2: Standard CityGML Levels of Detail (LoD) for building models, from LoD 0 (2.5D footprints) through LoD 4 (detailed architectural interiors).

2.4.3 Data sources and collection platforms

Geospatial data for urban analysis originates from a variety of sources, broadly categorized into remote sensing platforms and administrative/ancillary datasets.

Remote Sensing Platforms

Remote sensing, or *Earth Observation* (EO), involves collecting information about the Earth's surface without direct physical contact, primarily using satellite or airborne sensors. These platforms provide diverse forms of imagery and measurements (see Figure 2.3):

- **Aerial Imagery:** Captured from aircraft like drones, helicopters, or airplanes, aerial photography offers highly detailed visual information suitable for precise mapping and local to regional studies in urban planning and environmental monitoring.
- **Satellite Data:** Satellites provide comprehensive coverage of the Earth's surface, useful for terrain mapping, environmental change monitoring, and urban planning over extensive areas. They also support hyperspectral and multi-spectral imaging, capturing data across multiple spectral bands for detailed analysis of land and water features.
- **LiDAR (Light Detection and Ranging):** This laser-based technology precisely measures distances to the Earth's surface. Aerial LiDAR systems emit laser pulses, and by measuring the return time, they generate highly accurate elevation data, critical for creating detailed terrain models and 3D representations of buildings. For instance, Stuttgart uses airborne laser scanning combined with aerial photographs to derive building outlines.
- **Synthetic Aperture Radar (SAR):** An advanced radar technology, SAR creates detailed images of the Earth's surface. Its key advantage is the ability to penetrate clouds and collect data day or night, ensuring reliability in all weather conditions. SAR data is valuable for applications like surface deformation monitoring, flood mapping, and land cover classification.

Administrative and Ancillary Datasets

These sources provide structured, often tabular, information that complements remote sensing data by adding semantic depth:

- **Administrative Data:** This includes structured information about administrative boundaries (e.g., country borders, states, municipalities) and associated attributes like population statistics or electoral districts.

Geospatial Data Collection Platforms

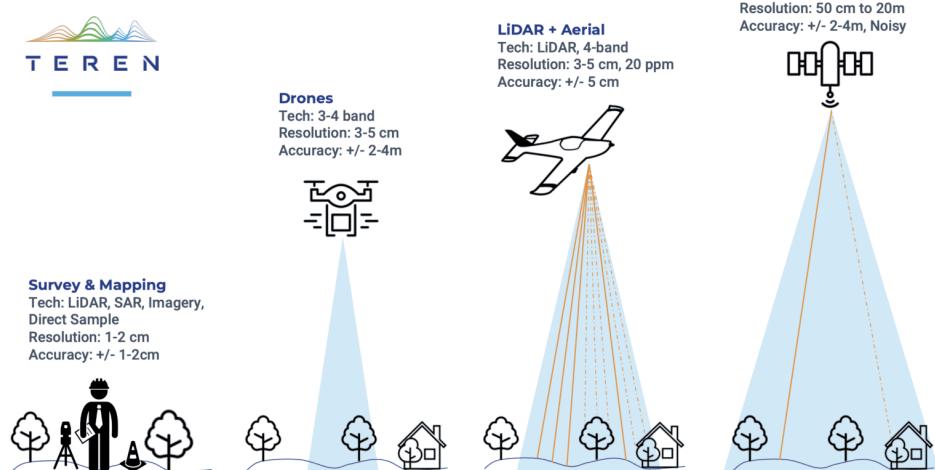


Fig. 2.3: Geospatial data collection platforms ranging from ground survey to drones, aerial LiDAR, and commercial satellites.

- **Ancillary Datasets:** Local authorities and utility providers maintain extensive catalogs of data, often in formats like shapefiles, covering information related to biodiversity, city development, electrical substations, and power cable networks. Crucially, these datasets provide rich tabular metadata linked to spatial features, such as building usage, type, age, number of storeys, date of refurbishment, and window proportion, which are vital for accurate analyses like heating demand calculations. Examples include national (e.g., Project Tabula in Germany) and regional (e.g., Bavaria, Schleswig-Holstein) data libraries.

2.4.4 Geospatial Data Formats

Raster Data: GeoTIFF for Aerial and Satellite Imagery

GeoTIFF (Geographic Tagged Image File Format) [Atl25] is a widely adopted meta-data standard that allows georeferencing information to be embedded directly within a TIFF file. This format is popular due to its capability to store high-quality raster images associated with satellite data, aerial photography, and scanned maps.

The “Geo” [Atl25] component of GeoTIFF specifically refers to the georeferencing metadata it contains, which assigns a real-world geographic location (coordinates) to each pixel in the raster image. This embedded geospatial information is crucial, as it allows users to accurately overlay these raster maps with vector data (e.g., highways,

towns, administrative borders), providing essential spatial context. GeoTIFF files often utilize a Projected Coordinate System (PCS), which employs linear units like meters or feet, facilitating precise spatial operations such as distance calculations. The integration of georeferencing information directly into the image data ensures that the geographic context is preserved within a single file, although this typically results in larger file sizes compared to standard TIFFs.

GeoTIFF is an OGC Implementation Standard and is in widespread use, particularly within NASA Earth science data systems, due to strong software support from open-source libraries (e.g., `libgeotiff`, `GDAL`, `Rasterio`) and commercial GIS products. For cloud-based computing environments, the use of Cloud Optimized GeoTIFF (COG) is recommended, as it enables efficient access while maintaining backward compatibility with existing tools. GeoTIFF serves as the standard format for the “Aerial Images” component of this thesis. Its georeferencing capability ensures that the visual data is spatially accurate and can be seamlessly integrated with other geospatial data types, such as vector building footprints or 3D models. This spatial alignment is fundamental for any AI model that needs to understand the real-world location and context of the physical structures it analyzes or generates [NAS25].

Vector and Tabular Data: GIS Shapefiles and Spatial Databases (PostGIS/PostgreSQL)

For managing and analyzing vector and tabular geospatial data, GIS shapefiles and spatial databases like PostGIS/PostgreSQL are indispensable. A Shapefile is a widely used geospatial vector data format designed specifically for Geographic Information System (GIS) software. It allows for the storage of vector features (points, lines, and polygons), where each feature can hold associated attribute data in a database format. This makes shapefiles a common format for administrative data, city development information, and utility network data maintained by local authorities and service providers [WWC16].

PostGIS is a powerful open-source extension to the PostgreSQL object-relational database system, enabling the creation of robust spatial databases. These spatial databases are designed for efficient management and organization of geospatial data, supporting both vector and raster formats. PostGIS offers comprehensive GIS functionality, including full OpenGIS support, advanced topological constructs (e.g., coverage, surfaces), and capabilities for raster and vector processing and analysis. It also facilitates spatial reprojection and provides robust import/export functionalities for formats like ESRI shapefiles. A key advantage of PostGIS is its

high interoperability with major GIS and Web GIS software packages, making it a popular choice due to its open-source nature, frequent updates, and rich feature set compared to proprietary alternatives [Jen15].

The city of Stuttgart, for instance, leverages PostgreSQL to directly store its CityGML data [Cit25], demonstrating the capability of spatial databases to house complex 3D city models. Tools like `shp2psql_gui` simplify the process of loading shapefiles into PostgreSQL, and desktop GIS software like QGIS can directly connect to and visualize data stored in PostGIS databases.

2.4.5 Visualization Techniques for Urban Geospatial Data

Effective visualization is paramount for interpreting complex geospatial datasets and communicating insights, particularly in urban planning and energy systems. This section explores common tools and approaches for rendering urban geospatial data, emphasizing their role in interactive analysis and decision support.

Geographic Information Systems (GIS) Tools: QGIS

Geographic Information Systems (GIS) [QGI25] software are fundamental tools designed to capture, store, manipulate, analyze, manage, and present all types of geographical data. Among these, QGIS stands out as a leading free and open-source GIS software, widely recognized for its extensive capabilities and active community development model.

QGIS offers exceptional cartographic design features, enabling users to create high-quality maps for various media, including desktop, mobile, cloud, and print. Its intuitive layout designer supports professional map production, and it includes powerful reporting tools for generating atlases with both maps and tabular content. The software provides unmatched capabilities for editing layers, with powerful digitizing and construction tools that allow for precise crafting and editing of points, lines, polygons, and meshes. It includes advanced construction tools for building geometries with curves and CAD-like precision, alongside highly customizable forms for user-friendly data input.

For data processing and analysis, QGIS integrates a comprehensive set of state-of-the-art analysis tools. Users can create automated analysis workflows by visually combining these tools, and the platform's extensible environment supports a diverse ecosystem of third-party analysis tools for expanded capabilities. QGIS ensures

industry-leading format support, addressing data integration challenges across various data sources and formats. It promotes standards and interoperability, allowing users to explore, utilize, and create web services based on industry standards, and to publish their work to cloud and mobile devices for wider accessibility.

Map-based and 3D Interactive Visualization for Urban Energy Systems

Visualization is a core functionality for interactive applications and systems built upon complex 3D city models. The objective is to provide high-quality, scalable, and cost-efficient rendering of massive 3D urban geometries and textures. Real-time rendering techniques are employed to achieve this, supporting various levels of abstraction from 2D map views to detailed 3D visualizations, enabling multiperspective interaction with 3D city models.

The “smart map” [Hoc24; Wik25a] approach exemplifies map-based visualization for 3D city models. This technique provides massive virtual 3D city models on various platforms (web browsers, smartphones, tablets) through an interactive map composed of artificial oblique image tiles. The 3D rendering is entirely server-side, generating pre-rendered map tiles for different levels of detail. This decoupling of data complexity from transfer complexity simplifies client implementation and enables scalable deployment for many concurrent users, without streaming the underlying complex 3D data to client devices.

A prominent example of this approach is the **UrbanTwin** [Fra25] platform developed by Fraunhofer IEE. UrbanTwin represents a comprehensive, high-resolution digital twin of urban areas, designed to support energy transition efforts at district and city scale. It combines building-level 3D data, behavioral and census statistics, technical infrastructure layers, and AI-enhanced insights into an interactive spatial decision-support environment. Key capabilities include:

- **High-resolution 3D building models** integrated with geospatial, demographic, and infrastructure data for energy demand and supply modeling.
- **Synthetic data generation using AI**, including time series and population profiles to fill data gaps in spatial energy modeling.
- **Simulation and visualization of energy scenarios** such as PV potential, heat demand, and EV infrastructure rollout in real-time 3D.
- **Stakeholder-inclusive interfaces** that facilitate communication between city planners, utilities, and the public through accessible, map-based tools.

- **Integration of aerial and satellite imagery** to assess rooftop characteristics, vegetation, and surface types relevant to heating or photovoltaic modeling.

UrbanTwin exemplifies the fusion of advanced geospatial visualization with AI-driven analytics and participatory planning. For this thesis, it demonstrates how a digital twin framework can act as a powerful medium for visualizing and interacting with AI-generated urban scenarios—bridging technical insights and stakeholder understanding in the context of sustainable urban energy systems.

The prevailing trend in urban geospatial visualization is toward highly scalable, interactive, and web-accessible 3D environments. This is crucial for Smart Cities initiatives where diverse stakeholders, including non-GIS experts, need to access and interact with complex urban data for informed decision-making. For Generative AI for data analysis of physical structures, this means that AI-generated scenarios or analyses (e.g., optimized solar potential maps, proposed retrofitting simulations) can be effectively communicated, explored, and validated by a broader audience, fostering collaborative planning and public engagement.

Methodology

” *Innovation distinguishes between a leader and a follower.*

— **Steve Jobs**
(CEO Apple Inc.)

The study began by defining its primary objective: to determine whether an attic was already used as living space or had the potential to become habitable. To support this, a shortlist of promising vision-language models was first identified by consulting public leaderboards and community rankings. Simultaneously, a focused literature review explored recent applications of these models in remote sensing and urban analysis, guiding decisions on how best to combine visual and tabular information, especially when distinguishing fine-grained roof details like dormers and skylights.

Next, the dataset was assembled. Decent-resolution aerial images of various roof type residential buildings in Kassel were retrieved from UrbanTwin and cross-checked against Google Earth to ensure clear visibility of dormers and skylights. Each image was then manually annotated to mark these features, which serve as key indicators of attic usability. Complementary building attributes (roof slope, roof height, and roof type) were extracted from a PostgreSQL database, cleaned, standardized, and linked to the annotated images. Finally, three evaluation tracks were executed: image-only (roof-feature detection), tabular-only (attribute-based prediction), and a combined multimodal approach. These tracks were used to quantify how blending visual cues with structured data improves attic-use classification.

All experiments were run on a GPU-equipped workstation, and system performance was measured using accuracy, precision, recall, and response latency to ensure the solution was both reliable and responsive.

3.1 Vision Language Model Selection

The goal of this stage was to identify a suitable Vision Language Model (VLM) capable of interpreting fine-grained features in aerial imagery, such as dormers and skylights, while balancing performance, resource efficiency, and deployment feasibility. A systematic multi-step process was followed—starting with an exploration of the VLM landscape, followed by public leaderboard analysis, model-wise comparisons, and practical feasibility checks. Figure 3.1 outlines the step-by-step process used to evaluate and select the vision language model for this study. This culminated in the selection of a single model, **Qwen-2.5-VL-7B-Instruct**, as the foundation for downstream multimodal inference tasks.

3.1.1 Step 1: Review of VLM Landscape

The Hugging Face blog on Vision Language Models (VLMs) [Fac24] served as an essential resource for the initial exploration of the VLM landscape. It offers a comprehensive overview of prominent open-source VLMs, along with curated comparisons that highlight differences in architecture, capabilities, and benchmark performance. The blog also provides practical guidance on how to evaluate and select suitable models using leaderboard ranking platforms such as LM Arena and OpenCompass, making it particularly useful for narrowing down candidates. Additionally, it includes technical insights into the modalities supported, inference behavior, and fine-grained image-text alignment performance, all of which are critical considerations for this study's multimodal analysis tasks.

3.1.2 Step 2: Consultation of Public Leaderboards

To obtain a comparative overview of candidate models, two public leaderboards were consulted. These platforms provided aggregated evaluation results across a range of vision-language tasks, enabling a quick assessment of model capabilities in zero- and few-shot settings:

- LM Arena [Are24] was utilized as a central platform for evaluating and comparing vision-language models (VLMs) in a community-driven and interactive setting. The general leaderboard (see Figure 3.2) provides ranked performance scores for a wide range of large language and multimodal models, based on over three million crowdsourced user votes. For VLM-specific evaluation, the

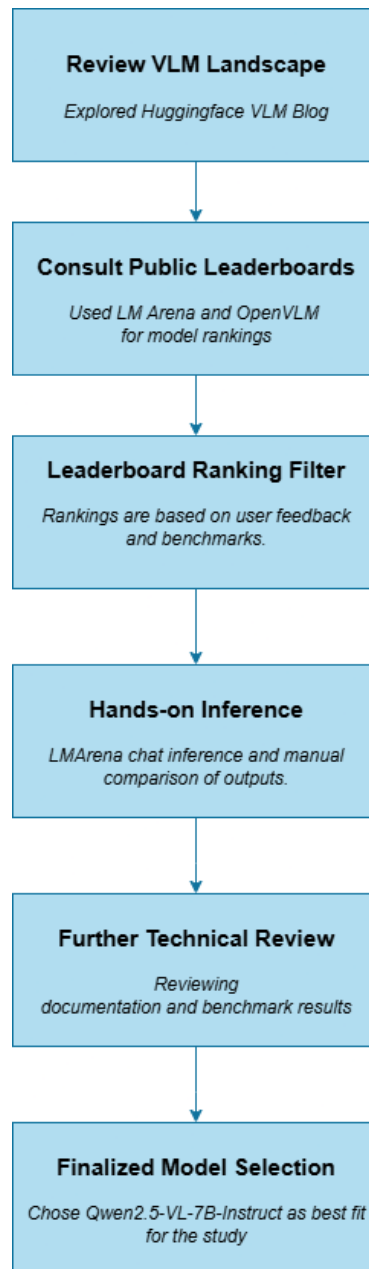


Fig. 3.1: Workflow for Vision Language Model (VLM) selection.

"Vision" tab (Figure 3.3) allows focused benchmarking on image-based tasks, helping identify models with superior image-text alignment and visual reasoning capabilities. In addition, LM Arena's "Direct Chat" playground (Figure 3.4) supports real-time multimodal inference, enabling direct model testing with both text and image inputs. This combination of quantitative comparison and qualitative interaction made LM Arena a valuable tool for filtering candidate models in the early selection phase.

- **OpenCompass "Open VLM Leaderboard"** on Hugging Face Spaces [Tea24a] – provided consolidated benchmark scores for over 270 open-source models across 31 multimodal datasets. The leaderboard interface supports advanced filtering by model size, type, and evaluation metric, allowing detailed comparison of mid-sized VLMs (4B–10B) on tasks such as image reasoning, OCR, and visual QA. While the current leaderboard (as of mid-2025) highlights newly released models at the top, it is important to note that during the early stages of this research (Jan–Feb 2025), **Qwen2.5-VL-7B-Instruct** was consistently ranked among the top-performing models. This temporal context was essential for its selection at that point in time (see Figures 3.5 and 3.6).

3.1.3 Step 3: Model Shortlisting Based on Leaderboard Filtering

Based on the insights obtained from public leaderboards, an initial shortlisting of models was performed by prioritizing those consistently ranked among the top performers in vision-language tasks. The LM Arena and OpenCompass leaderboards provided both community-driven scores and benchmark-based rankings across key multimodal evaluation datasets.

The following six models emerged as top contenders from this leaderboard-based filtering process: **Pixtral-Large-Instruct-2411**, **Llama-3.2-90B-Vision-Instruct**, **Pixtral-12B-2409**, **Qwen-2-VL-7B-Instruct**, **Qwen-2.5-VL-7B-Instruct**, and **LLaVA-v1.6-34B**.

3.1.4 Step 4: Hands-on Inference-Based Evaluation

After leaderboard-based shortlisting, the selected models were subjected to hands-on inference using LM Arena's Direct Chat interface. This practical evaluation involved testing each model on a **complex aerial image** containing multiple object

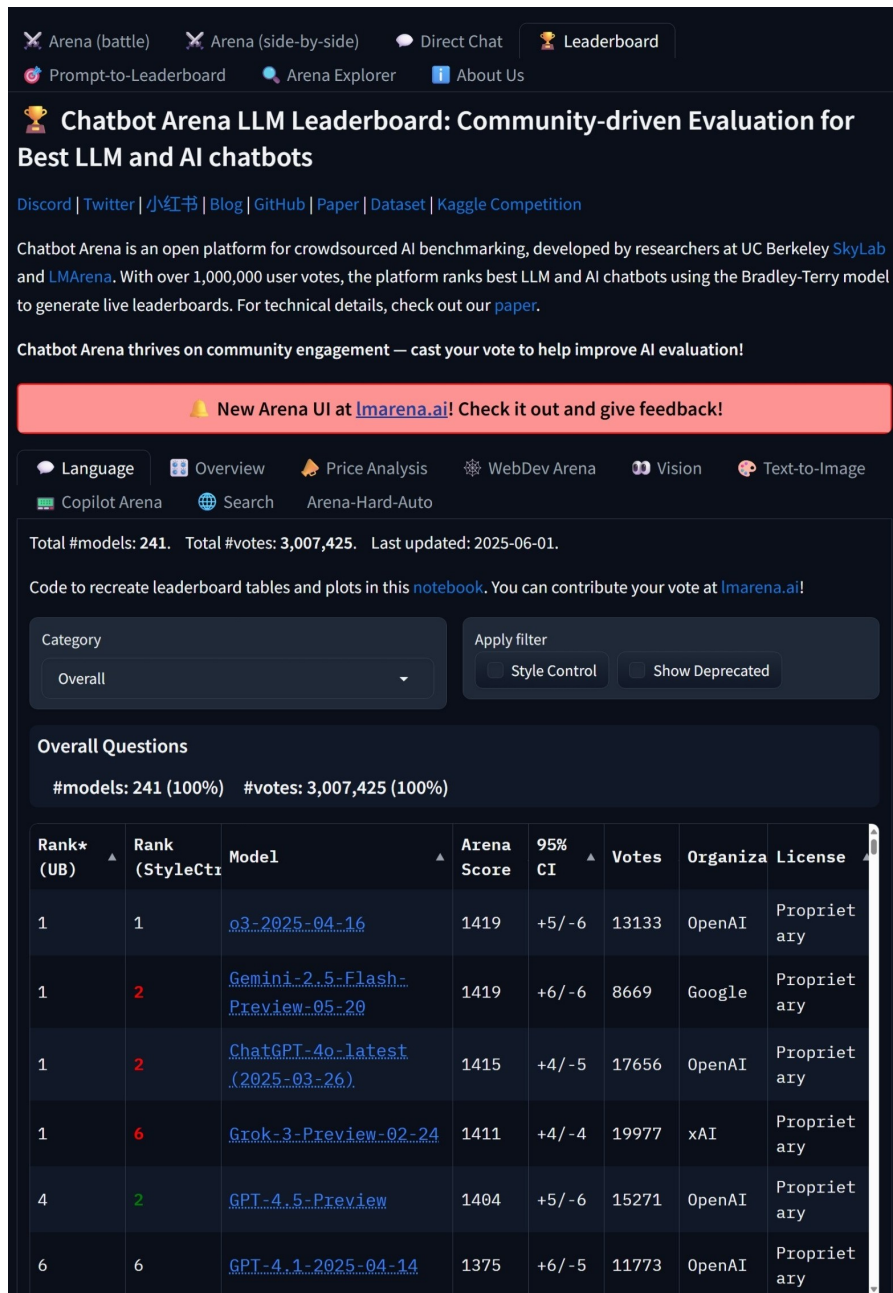


Fig. 3.2: General leaderboard on LM Arena showcasing overall rankings across language and multimodal models.

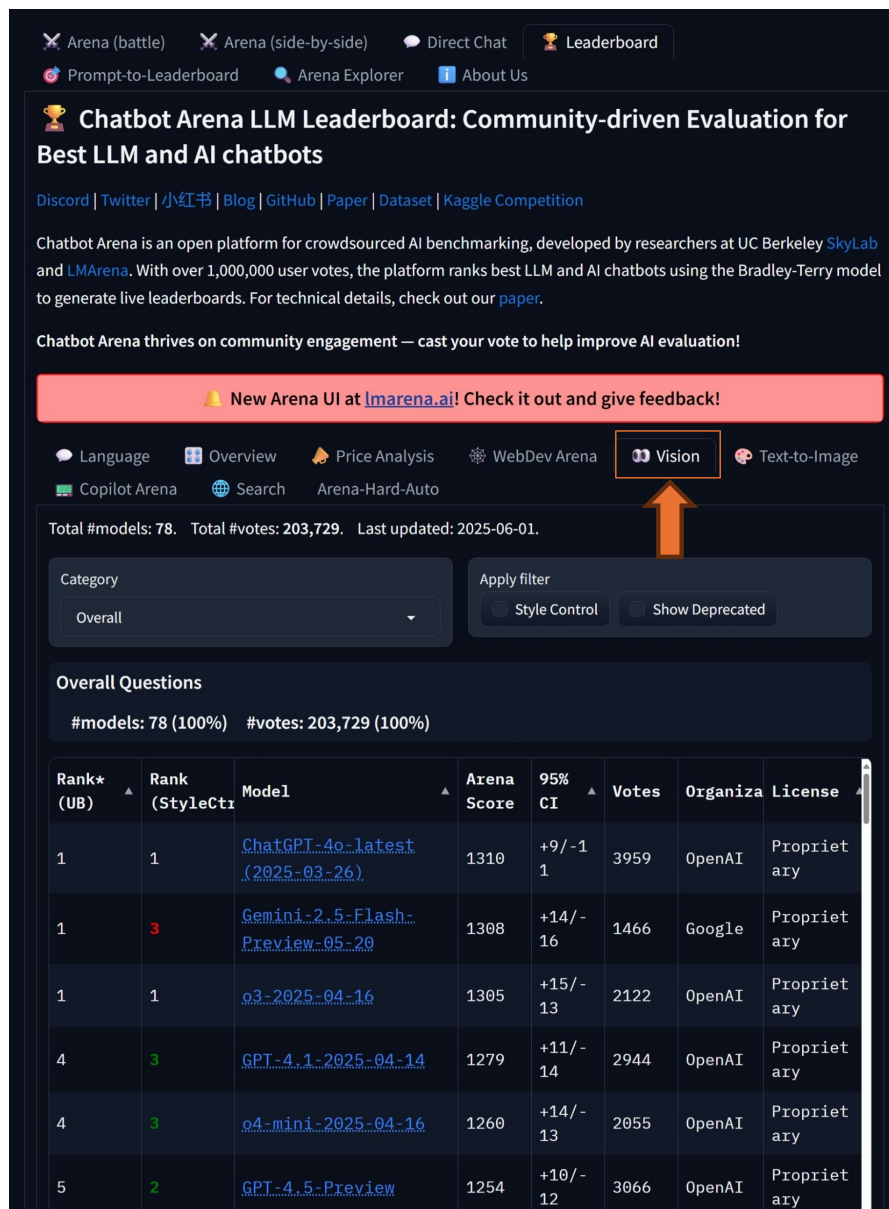


Fig. 3.3: Vision-specific leaderboard on LM Arena used to compare performance across vision-language models.

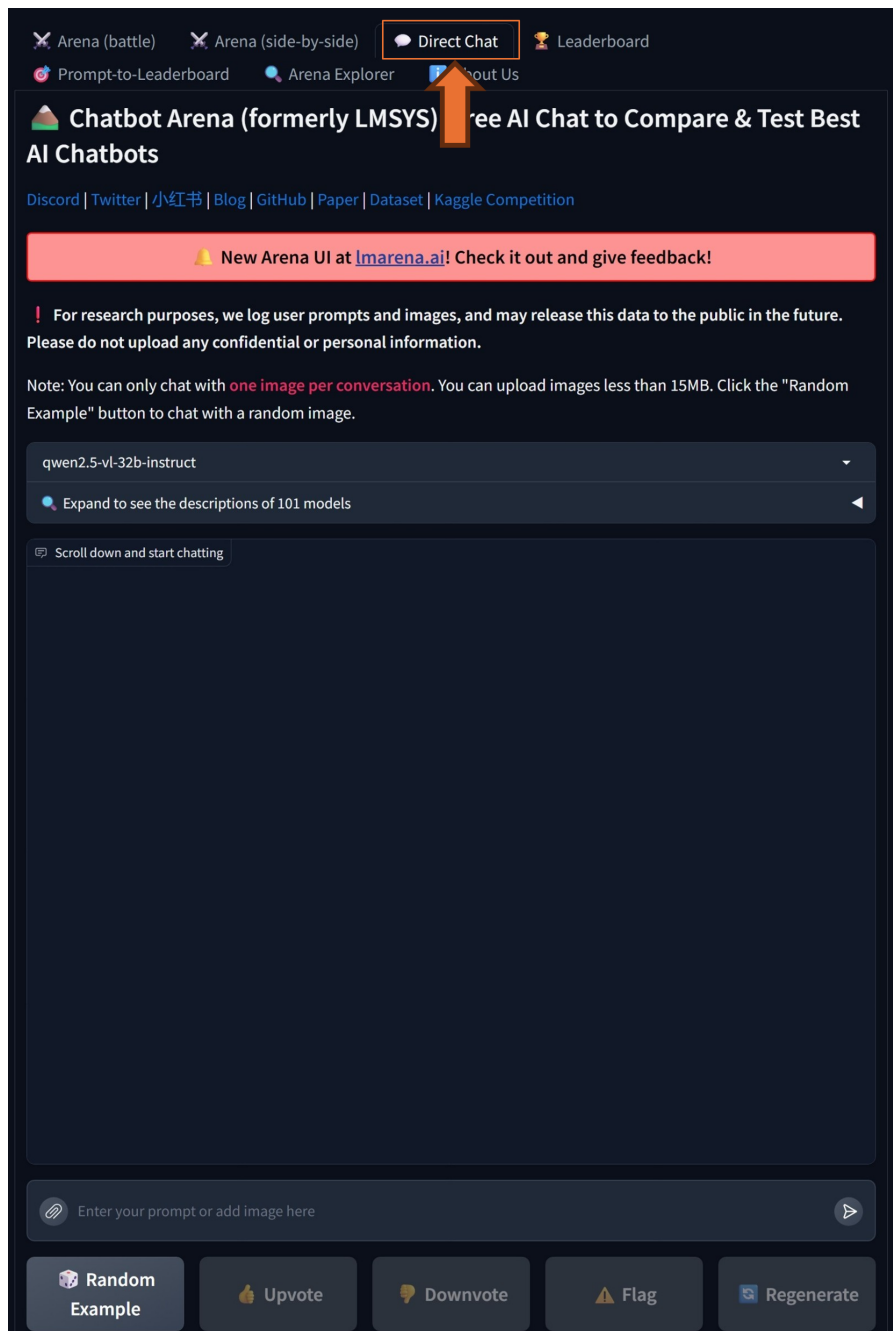


Fig. 3.4: Interactive Direct Chat interface of LM Arena, allowing users to test models with image and text input.



Fig. 3.5: Homepage of the Open VLM Leaderboard showing available benchmarks and evaluation criteria.

Evaluation Dimension

☒ Avg Score

☒ Avg Rank

☒ MMBench_V11

☒ MMStar

☐ MME

☒ MMMU_VAL

☒ MathVista

☒ OCRBench

☒ AI2D

☒ HallusionBench

☐ SEEDBench_IMG

☒ MMVet

☐ LLaVABench

☐ CCBench

☐ RealWorldQA

☐ POPE

☐ ScienceQA_TEST

☐ SEEDBench2_Plus

☐ MMT-Bench_VAL

☐ BLINK

Model Name

Input the Model Name (fuzzy, case insensitive)

Model Size

☐ <4B

☒ 4B-10B

☐ 10B-20B

☐ 20B-40B

☐ >40B

☐ Unknown

Model Type

☐ API

☒ OpenSource

Rank	Method	Param (B)	Language Model	Vision Model	Eval Date	Avg Score	Avg Rank	MMBench_V11	MMStar	
1	SAIL-VL-1.6-8B	8.33	Qwen2.5-7B	AIMv2 Huge	2025/04/24	73.6	30.25	84	69.5	5
2	InternVL 3-8B	7.94	Qwen2.5-7B	InternViT-300M-v2.5	2025/04/14	73.6	38.38	82.1	68.7	6
3	InternVL 3-9B	9.14	InternLM3-8B	InternViT-300M-v2.5	2025/04/14	72.6	40.88	82.2	67.4	5
4	Ola-7b	8.88	Qwen2.5-7B	Oryx-ViT	2025/02/05	72.6	40.62	84.3	70.8	5
5	WeThink-Qwen2.5VL-7B	8.29	Qwen2.5-7B	QwenViT	2025/05/26	72.5	41.38	81.8	64.2	6
6	SAIL-VL-1.5-8B	7.95	Qwen2.5-7B	InternViT-300B	2025/04/23	72.4	37.5	82.1	69.7	5

Fig. 3.6: Filtered view showing benchmark scores for 4B–10B open-source models, highlighting Qwen2.5-VL-7B.

types—such as buildings, cars, trees, and parking areas—using the following three prompts:

- *Give me a detailed description of the image.*
- *Give me a detailed description of each individual object in the image.*
- *Give the details of the position of each of the objects in the image.*

This interactive process provided qualitative insights into each model’s descriptive power, spatial understanding, and robustness under real-world image complexity. The following key observations were made:

- **Level of Detail:** Larger models (e.g., Pixtral-Large, LLaVA-v1.6-34B, Llama-3.2-90B) offered more comprehensive and structured descriptions.
- **Spatial Awareness:** Pixtral-Large, Qwen-2-VL, and Qwen-2.5-VL-7B-Instruct demonstrated strong capabilities in localizing and contextualizing objects.
- **Handling Low Resolution:** Models like Llama-3.2-90B reported difficulty in interpreting low-resolution images.
- **Domain Emphasis:** Pixtral-12B focused more on urban structures, while Qwen-based models provided a broader perspective, capturing contextual and environmental cues.

3.1.5 Step 5: Detailed Review of Candidate Models

Each shortlisted model was then reviewed in detail using its technical documentation, benchmark results, and user-reported feedback. Specific attention was given to:

- The **Qwen2.5-VL-7B-Instruct** model page [Tea24b], which showcased improvements in instruction tuning and multi-object image interpretation.
- The **InternVL evaluation suite** [Tea24a], which helped assess model robustness on complex referential and spatial tasks, confirming Qwen2.5’s competitive standing relative to much larger models (see Figure 3.7).

3.1.6 Step 6: Finalized Model Selection

Three primary constraints informed the final model choice:

- **Compute feasibility:** LLaVA-v1.6-34B, Llama-3.2-90B and Pixtral-Large was excluded due to high computational demands and resources.
- **Regional accessibility:** The Llama-3.2-90B API was also found to be regionally restricted in EU zones (Figure 3.8), limiting practical deployability.
- **Quality vs. Size trade-off:** Qwen-2.5-VL-7B-Instruct offered robust spatial reasoning, detailed descriptions, and instruction-following capabilities within a mid-sized parameter range.

Qwen-2.5-VL-7B-Instruct was therefore selected as the primary VLM for all subsequent multimodal inference experiments, including spatial reasoning, object detection, and prompt-based analysis (see Section 3.4).

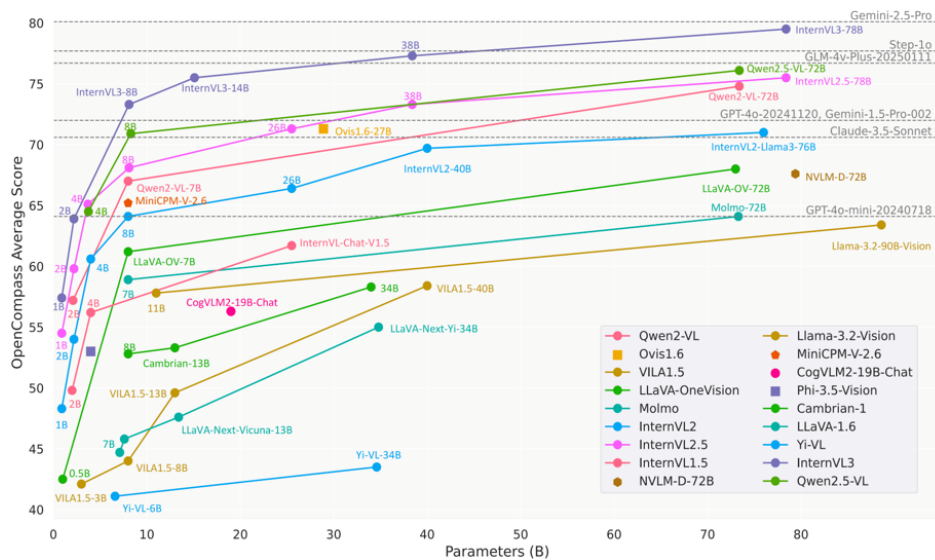


Fig. 3.7: OpenCompass leaderboard visualization of average benchmark scores versus model size (in billions of parameters). Qwen2.5-VL-7B is shown to perform competitively in the 7–8B parameter range [Ope24].

3.2 Data Preparation

Data preparation involved sourcing, cleaning, and aligning aerial imagery with detailed tabular metadata. The following subsections describe the origins of each

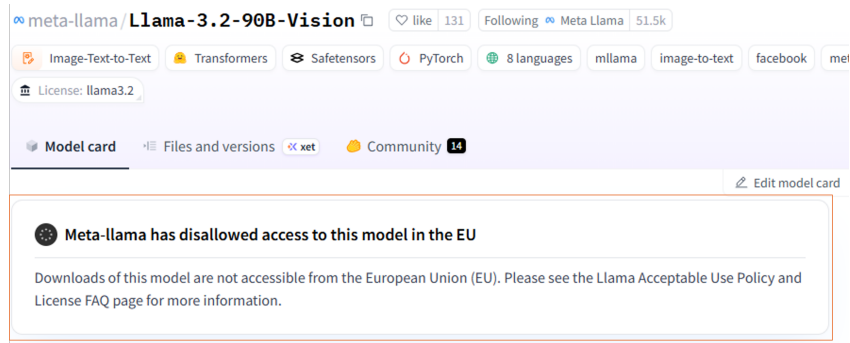


Fig. 3.8: Access restriction notice on the Hugging Face model page for Llama-3.2-90B-Vision. The model is not downloadable from the European Union.

modality, the preprocessing and annotation workflows, and the final integration steps.

3.2.1 Aerial Imagery Overview

- **Source:** UrbanTwin platform (Fraunhofer IEE) [Fra25], a 3D visualization tool based on the 2022 LOD2 building stock for Kassel. Figure 3.9 shows the tool interface.
- **Roof geometries included:** Flat, Pitched, Gable, Hip, Mansard, Tent, Mixed, Dome, and Shed roofs [Wik25b].
- **Building categorization:**
 - IWU schema (TABULA project) [Ins25]: Residential (SFH, MFH, TH) and Non-residential (offices, hospitals, industrial, supermarkets).
 - ALKIS cadastral classes [Adv25]: Residential, Commercial/Industrial, Public.
- **Annotation workflow:**
 1. Export aerial image tile for each LOD2 model from the UrbanTwin viewer. Figure 3.10 shows the building image present in the left panel.
 2. Cross-check each tile against Google Earth to verify dormers and skylights.
 3. Manually annotate dormers and skylights to produce bounding box labels.
- **Focus:** Residential buildings with sloped roofs (gable and hipped types).



Fig. 3.9: UrbanTwin opening interface



Fig. 3.10: Aerial Image of the building present on the information panel on the left

3.2.2 Tabular Metadata Overview

Each building model—identified by `Gid` (matching the image filename)—was linked to a PostgreSQL table containing. The same data can be seen in UrbanTwin in the Figure 3.11 and overview is there in 3.1 :

Tab. 3.1: Building Attribute Mapping from PostgreSQL Table

Attribute	Description
<code>Gid</code>	Unique building identifier (matches image filename)
Floors	Number of storeys (from <i>Etagen</i>)
<code>Roof_type</code>	Roof type code (3100 = gabled, 3200 = hipped)
Slope	Roof inclination angle (degrees)
<code>Roof_height_max</code>	Maximum roof height (meters)
Nutzfläche	Usable horizontal floor area (m ²)
Einwohner	Number of occupants
<code>Iwu_gebttyp</code>	IWU building subtype (e.g., <code>EFH_D</code> , <code>MFH_C</code>)
<code>Roof_area</code>	Total surface area of roof faces (m ²)
<code>Roof_volume</code>	Volume under the roof (m ³)
<code>Cube_volume</code>	Total building envelope volume (m ³)
Strasse	Street address



Fig. 3.11: Tabular data corresponding to the selected building can be seen on the left

3.2.3 Aerial Imagery Processing and Annotation

The aerial imagery underwent the following processing steps to prepare it for annotation and model training:

- **Selection of residential sloped-roof images:** Only tiles depicting residential buildings with flat, gable or hipped roofs were retained to focus on the target use case.
- **Removal of unsuitable structures:** Buildings with overly complex roof geometries or misaligned imagery artifacts were excluded to avoid annotation inconsistencies.
- **Feature identification:** Dormers and skylights were defined as the primary roof features of interest, serving as indicators of attic usability.
- **Annotation tool usage:** The MakeSense.ai platform [Mak25] was employed to label each image with "dormer" and "skylight" bounding boxes.
- **Cross-verification with Google Earth:** Due to resolution limitations in UrbanTwin imagery, Google Earth views were consulted to ensure accurate placement of annotations.

3.2.4 Tabular Metadata Extraction and Cleaning

- **Data source:** Records stored across multiple materialized views and tables in a PostgreSQL database.
- **Extraction method:** SQL queries linked each record to its aerial image via the common identifier (Gid).
- **Data provenance:** Core dataset provided by the Bundesamt für Kartographie und Geodäsie (BKG) [Bun25], ensuring authoritative geospatial information.
- **Attribute selection:** Only fields essential to attic-use classification—such as roof type, slope, roof height max, roof area, and roof volume were retained; non-essential columns were dropped.

3.2.5 Ground Truth Label Generation

To evaluate the capability of multimodal models in predicting attic usability, ground truth labels were constructed from two distinct perspectives: direct image interpretation and structured tabular reasoning.

Ground Truth Generation for Image Only Inference

The ground truth in case of 4.1 primarily is generated independent of the consideration of tabular metadata. This was to experiment if the model is able to make prediction based on just visual cues.

The following labels were manually assigned based on image inspection: Roof features - dormers, skylights, vents and roof type for binary classification for gable and hipped roof types; and attic usability - whether the attic appeared livable or had potential for habitation, without referring to tabular data.

Ground Truth Generation Based on Tabular Data Reasoning and Combination with Imagery

- **Minimum Ceiling Height Requirement:** According to building standards, habitable attic space requires a ceiling height of at least 2 m over half the net floor area (actual standard: 2.2 m).
- **Roof Height Threshold:** To ensure sufficient headroom, the maximum roof height (`roof_height_max`) was set to ≥ 4 m (twice the minimum ceiling height).
- **Roof Slope Considerations:** Comfortable living spaces typically require roof slopes above 35° , with 45° being common and well-suited. For this study, a conservative threshold of 30° was chosen.
- **Roof insights from images:** The presence of a dormer was criteria selected for generating the ground truth from the analysis of images. Dormer presence is a strong indication for the presence of habitable attic.
- **Ground Truth Cases:** To evaluate attic usability and conversion potential, two distinct ground truth definitions were established based on tabular attributes and visual cues. These were used across both tabular-only and image-tabular multimodal inference settings.

Case 1 (Actual Attic Usage Prediction):

This stricter condition reflects current attic usage as livable space. A building satisfies this case if it either has both a sufficiently steep roof slope ($\geq 30^\circ$) and a tall roof (≥ 4 m), or if a dormer is visually detected. Formally:

$$(\text{slope} \geq 30^\circ \wedge \text{roof_height_max} \geq 4 \text{ m}) \vee (\text{dormer} = 1)$$

Case 2 (Potential Conversion Exploration):

This more inclusive condition represents buildings with potential for future attic conversion. It is satisfied if either the slope or roof height meets the threshold, or if a dormer is present:

$$(\text{slope} \geq 30^\circ \vee \text{roof_height_max} \geq 4 \text{ m}) \vee (\text{dormer} = 1)$$

These rule-based ground truth labels served as the benchmark for evaluating the model's ability to infer attic usability using only tabular data, as well as when combining tabular metadata with visual features from aerial imagery.

3.3 Exploratory Analysis of Prepared Data

This section presents an exploratory analysis of key roof-related attributes extracted from the cleaned dataset used for 4.2 and 4.3. Specifically, we examine the distributions of roof heights and roof slopes for the two most prevalent roof types in the dataset- gable and hipped roofs. A total of 745 images were considered in which 375 were of gable type and 370 were of hipped type. These parameters are critical in evaluating attic livability potential, as minimum height and slope thresholds strongly influence usable attic volume. The following subsections visualize and interpret the spread of these attributes using histograms and pie charts, enabling a better understanding of structural constraints across the building stock.

3.3.1 Distribution of buildings

Distribution of Roof Heights for Gable Roofs

- The bar chart in Figure 3.12 shows that most gable-roofed buildings have a maximum roof height between 2 m and 4 m, with a clear mode around 3 m.

- A dashed vertical line at 4 m marks the minimum ceiling-height requirement for habitable attic space.
- Only a small tail of gable roofs exceed 4 m, indicating that naturally sufficient headroom is uncommon.
- The bar chart and the pie chart in Figure 3.13 together show that approximately 25 % of gable roofs meet or exceed the 4 m threshold, while 75 % fall below.
- This distribution implies that most gable-type residences would need structural modifications such as dormers or roof-raising to support a comfortable attic living space.

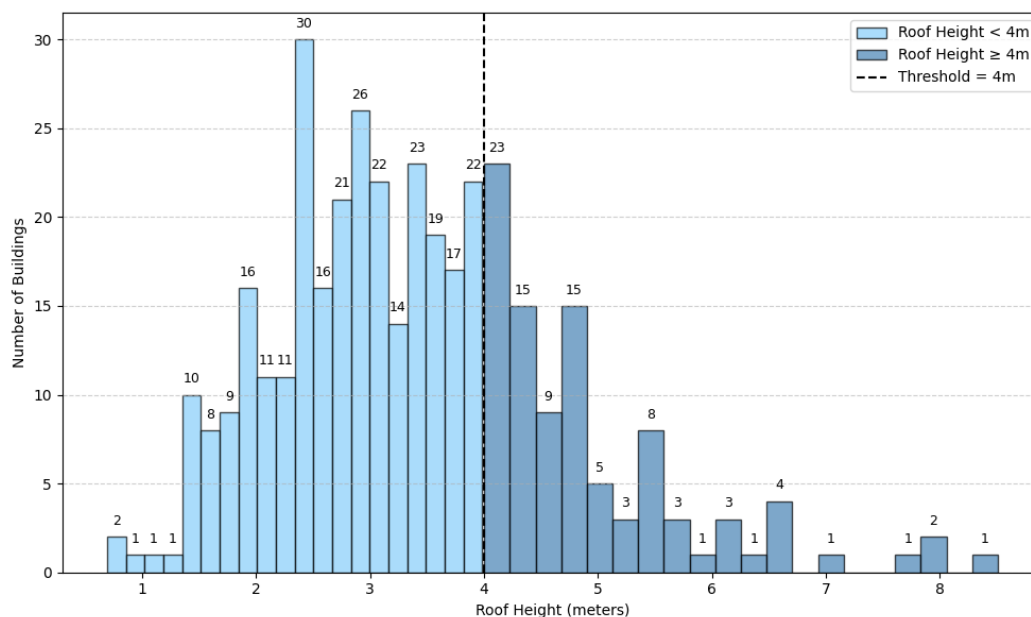


Fig. 3.12: Distribution of roof heights for gable-roofed buildings. A threshold of 4 m (dashed line) is marked as the minimum requirement for habitable attic space.

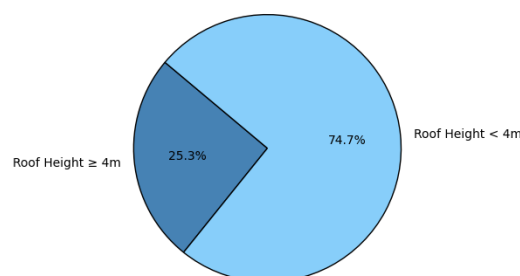


Fig. 3.13: Proportion of gable-roofed buildings by roof height relative to the 4 m attic habitability threshold.

Distribution of Roof Slopes for Gable Roofs

- The bar chart in Figure 3.14 shows that roof slope angles for gable-roofed residences range from approximately 10° to over 60° , with most values between 20° and 45° .
- A dashed vertical line at 30° highlights the minimum slope threshold chosen for habitable attic space.
- In the histogram (Figure 3.14), bars to the left of the threshold represent gable roofs with slope $< 30^\circ$, and bars to the right represent slope $\geq 30^\circ$.
- The pie chart in Figure 3.15 shows that about 34.7 % of gable roofs fall below 30° , while 65.3 % meet or exceed the threshold.
- The highest frequencies occur around 30° – 35° , indicating that many gable roofs are designed with moderately steep pitches.
- Overall, most gable roofs satisfy the minimum slope requirement for comfortable attic living, but a notable fraction would require pitch adjustments to be fully habitable.

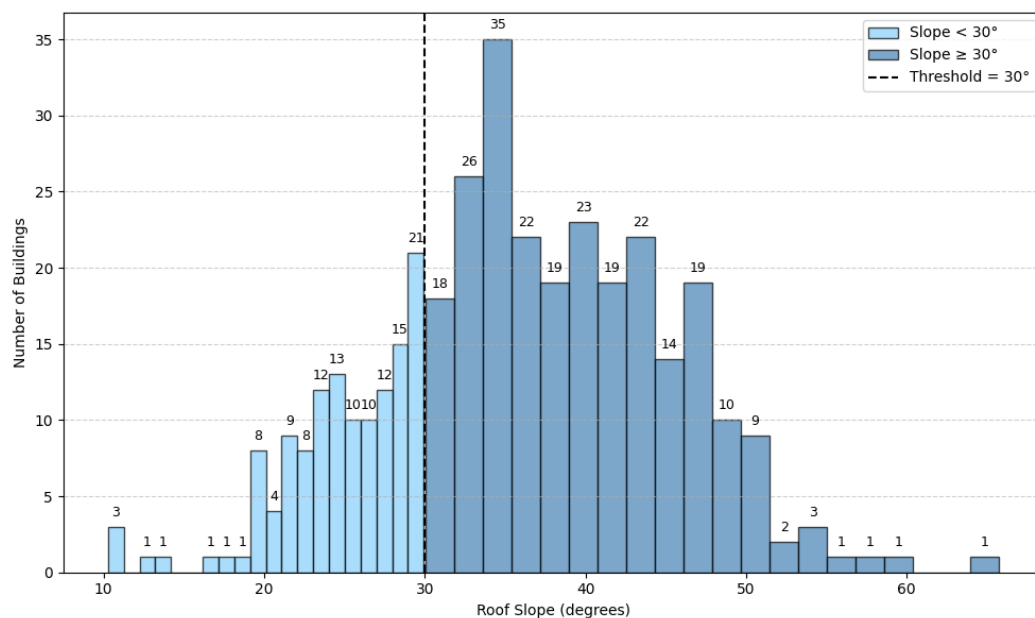


Fig. 3.14: Distribution of roof slopes for gable-roofed buildings. A vertical dashed line marks the 30° threshold for attic habitability.

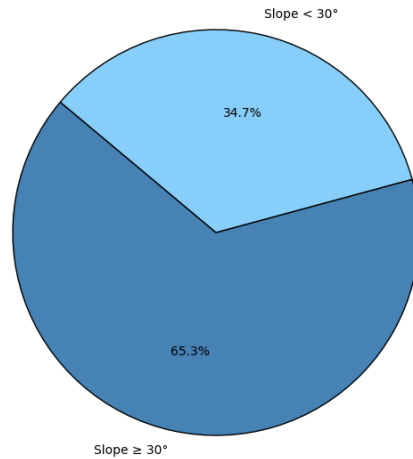


Fig. 3.15: Proportion of gable-roofed buildings with slopes above and below the 30° habitability threshold.

Distribution of Roof Heights for Hipped Roofs

- The bar chart in Figure 3.16 shows that hipped-roof buildings exhibit maximum heights ranging from about 1 m up to over 12 m, with the majority clustered between 2 m and 5 m.
- A dashed vertical line at 4 m marks the minimum headroom threshold for habitable attic space.
- In the histogram (Figure 3.16), bars left of the threshold represent heights < 4 m, while bars right of the line represent ≥ 4 m.
- The pie chart in Figure 3.17 shows approximately 27.8 % of hipped roofs meet or exceed 4 m, while 72.2 % fall below the threshold.
- These results indicate that only a minority of hipped-roof residences naturally satisfy the ceiling-height requirement, suggesting many would need structural adaptations (e.g., dormers or roof raising) to support a comfortable attic living space.

Distribution of Roof Slopes for Hipped Roofs

- The bar chart in Figure 3.18 shows that hipped-roof slope angles span from approximately 5° up to over 60°, with most values concentrated between 10° and 30°.

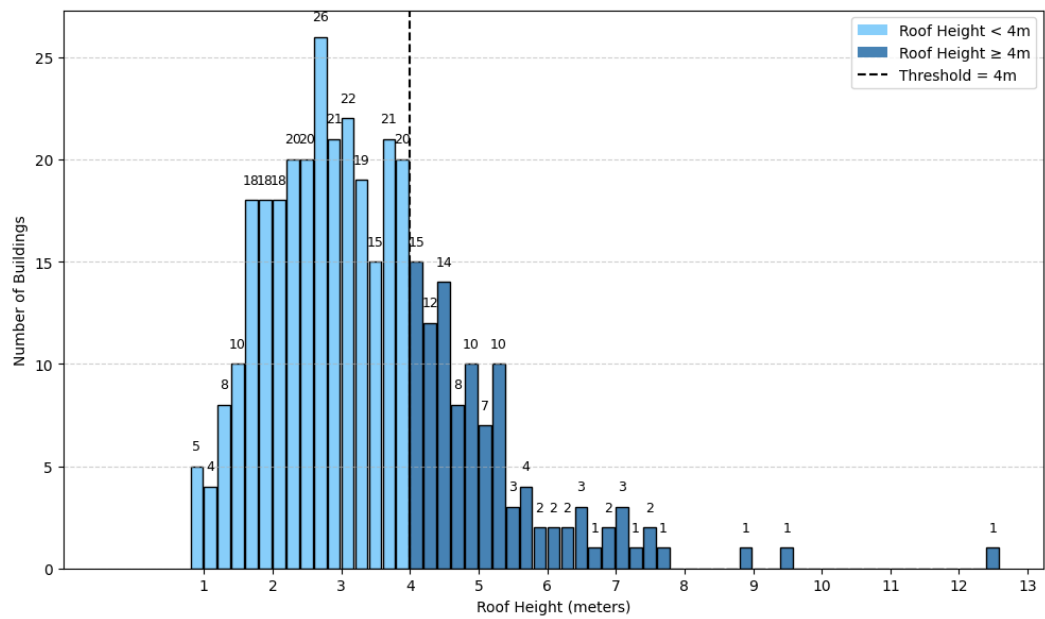


Fig. 3.16: Distribution of roof heights for hipped-roof buildings. A dashed line marks the 4 m threshold for habitable attic headroom.

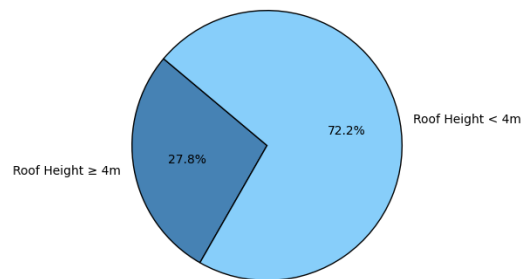


Fig. 3.17: Proportion of hipped-roofed buildings with roof height above and below the 4 m threshold.

- A dashed vertical line at 30° marks the minimum slope threshold chosen for habitable attic space.
- In the histogram (Figure 3.18), bars to the left of the threshold represent slopes $< 30^\circ$, while those to the right represent slopes $\geq 30^\circ$.
- The pie chart in Figure 3.19 shows that about 76.2% of hipped roofs have slopes below 30° , and 23.8% meet or exceed the threshold.
- Peaks in the histogram occur around 20° – 25° , indicating a common moderate pitch among hipped roofs.
- Overall, most hipped roofs fall short of the minimum slope requirement for comfortable attic living, suggesting many would benefit from pitch adjustments or dormer additions.

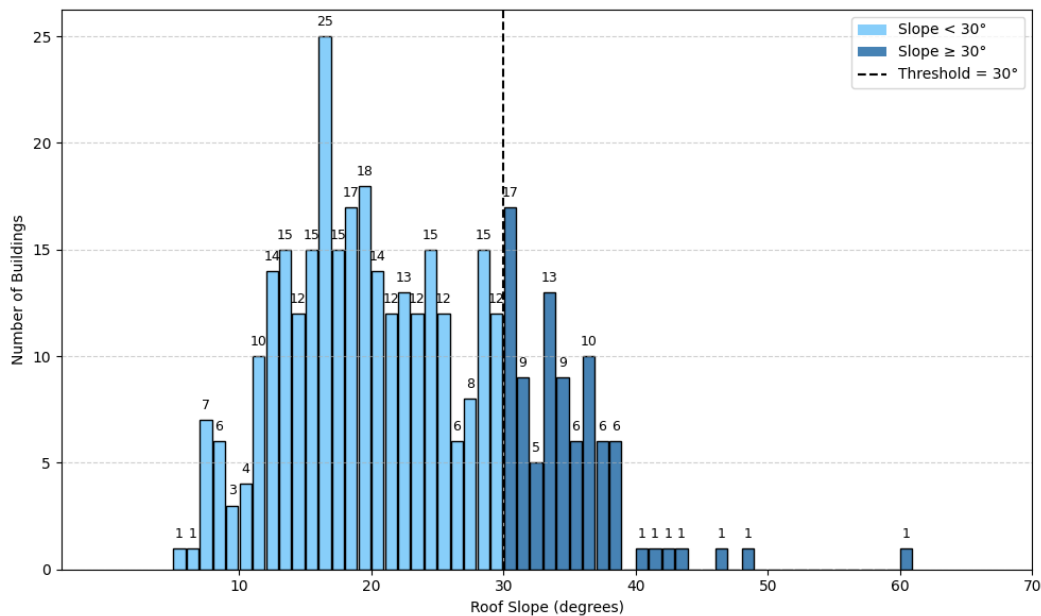


Fig. 3.18: Distribution of roof slopes for hipped-roof buildings. A dashed vertical line at 30° marks the attic habitability threshold.

3.4 Evaluation Framework

This research adopts a structured evaluation framework to assess the performance of open-source Vision-Language Models (VLMs) on the task of analyzing rooftop features from aerial images and associated tabular metadata. The framework is designed to address all three research questions and includes both single-modality

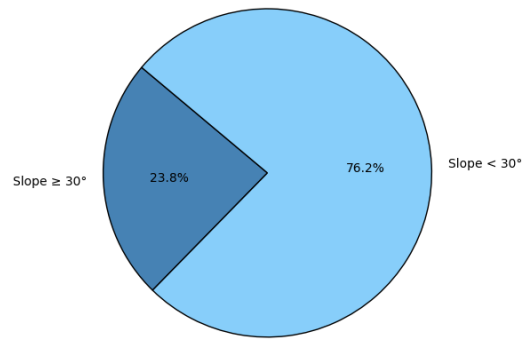


Fig. 3.19: Proportion of hipped-roofed buildings with slopes above and below the 30° threshold.

(image-only 4.1 and tabular-only 4.2) and multimodal (image-plus-tabular 4.3) inference settings. Each model's output is compared against predefined ground truth conditions derived from expert-defined criteria. Standard classification metrics such as accuracy, precision, recall, and F1-score are used to quantitatively evaluate performance. This framework enables systematic comparison across inference modes, highlighting the strengths and limitations of VLMs in geospatial analysis tasks.

3.5 Conclusion

The methodology presented in this chapter established a rigorous, workflow for predicting attic habitability in Kassel's residential building stock. A systematic literature review and leaderboard survey grounded the selection of state-of-the-art vision-language models, culminating in the choice of Qwen-2.5-VL-7B-Instruct under practical constraints of compute and regional availability. Aerial imagery was then procured from UrbanTwin and meticulously annotated cross-checked with Google Earth to capture dormers and skylights, while tabular metadata from a PostgreSQL database provided complementary structural attributes. These two modalities were aligned via a common file-name identifier, cleaned, and split into training, validation, and test sets to ensure balanced representation of gable and hipped roofs as well as attic-use labels.

A comprehensive exploratory analysis clarified the distributions and interrelationships of roof geometries and building features, informing feature selection and threshold definitions for ground truth. The evaluation framework that followed

leveraged zero and few-shot prompting, role-based system messages, and confusion-matrix metrics across image-only, table-only, and combined inference tracks. By comparing uni-modal baselines against the integrated multimodal pipeline, the methodology quantified the value of fusing visual annotations with structured data. Together, these steps have laid a robust foundation for the results chapters, demonstrating both the feasibility and the added accuracy of a multimodal approach to attic-use classification.

Evaluation and Results

“Users do not care about what is inside the box,
as long as the box does what they need done.

— **Jef Raskin**

about Human Computer Interfaces

This chapter presents a comprehensive evaluation of the proposed attic space detection methods across three complementary inference paradigms: vision only, tabular only, and a combined images with tabular approach, using Qwen2.5 VL 7B Instruct. In the first set of experiments, zero shot prompts of varying specificity and system roles (“Helpful Assistant” vs. “Expert”) are applied to aerial roof imagery drawn from two datasets (mixed sloped with flat, and exclusively sloped) to gauge raw visual reasoning. Few shot strategy was also employed on the best prompt in combination of best system role. Next, purely tabular inference rules based on literature research derived feature thresholds (`slope_max` and `roof_height_max`) are assessed on gable, hipped, and combined roof subsets. Finally, a hybrid strategy augments these numeric rules with an explicit dormer detection prompt to reconcile geometric and visual cues. For each scenario, confusion matrices and standard metrics (accuracy, precision, recall, F1 score) are reported, highlighting the trade offs between strict constraint enforcement and broader candidate recall. The following sections detail these results and analyze how prompt design, persona framing, and modality integration influence model performance.

4.1 Prompt-Based Inference Utilizing Only Imagery

This section presents the core inference task determining whether an aerial roof image indicates habitable attic space through two complementary cases. First, we treat it as a straightforward binary decision problem (Experiment 1), in which the model simply answers “Yes” or “No” based on its learned image representations. Next, in (Experiment 2), the model is tasked with five independent binary classification sub tasks, each targeting a particular roof attribute. For each aerial image, the

model must answer “Yes” or “No” by properly analyzing visual cues. Each prompt is designed to focus the model’s attention on distinct architectural features such as skylights, vents or chimneys, and overall roof geometry, enabling a fine grained evaluation of its zero shot visual reasoning capabilities.

4.1.1 Experiment 1 - Aerial Roof Attic Utilization: Framing as Binary Classification with Yes/No VQA

In this experiment setting, the model’s baseline capacity to infer attic living space is evaluated solely from visual input and a minimal yes/no prompt. This was performed to evaluate the model performance across image complexities. In this experiment the dataset-1 consists of combination of flat roofs along with sloped roofs. With the presence of flat roofs which is simplest in complexity, the model is expected to perform better than in the case of dataset-2 which excludes flat roofs and only includes sloped roofs which are complex in nature and makes it difficult for the model to predict. The difference in attic utilization for flat and sloped roofs is that, there is no attic in flat roofs which is not true in the case for sloped roofs. Three distinct zero-shot prompts P1, P2, and P3 are employed, each phrased slightly differently but all requesting a “Yes” or “No” response regarding the presence of habitable space under the roof. The zero shot prompts are tuned in such a way that the model is expected to perform with increasing accuracy. This setup assesses how effectively the pre trained vision language model leverages its learned representations such as roof pitch, dormer like structures, and volumetric shadows to make a binary decision across diverse roof types without fine tuning or specialized guidance. Metrics recorded accuracy, precision, recall, and F1 score establish the reference performance. After this process the few-shot prompting is applied on the case with best accuracy for dataset-2 to check if it improves accuracy in case of complex images in comparison to zero-shot prompting.

Datasets

- **Dataset-1:** It consists of combination of flat and sloped roofs. Total 200 images are present. This dataset is of lower complexity due to presence of flat roofs.
- **Dataset-2** – It consists of only sloped roofs and excludes the flat roofs. Total 201 images are present. This dataset is of higher complexity due to absence of flat roofs and sloped roofs are architecturally more complex.

System Roles

- **Role 1:** “Helpful assistant” (generic instruction)
- **Role 2:** “Expert in analyzing geospatial aerial images of residential buildings, including roof classification, roof volume analysis, and determination of roof-space utilization.”

Prompting Strategies

Two prompting strategies are evaluated:

1. **Zero-Shot Prompting:** Three distinct prompts (P1–P3) are issued without any example image–label pairs.
2. **Few-Shot Prompting:** A small set of 4 input-output pairs are taken, where input consists of image + prompt and output consists of text output(Yes/No).

Zero Shot Prompt Strategy

In this strategy zero shot prompting was performed using three prompts for the two system roles mentioned above to evaluate the impact of both system roles and prompts. Total six evaluation cases were obtained for which this strategy has been performed.

- **P1:** *“Analyze the roof of the building in the image and determine whether the volume of the roof is utilized as living space. Answer ‘Yes’ if the roof indicates the presence of living space; otherwise, answer ‘No.’”*
- **P2:** *“Analyze the image and determine if there is living space under the roof of the building. Consider roof features like slope, dormer windows, skylights, and overall roof volume. Answer ‘Yes’ if the roof design suggests the presence of living space; otherwise, answer ‘No.’”*
- **P3:** *“Analyze the image and determine if there is living space under the roof of the building. Consider roof features like slope, dormer windows, skylights, and overall roof volume. Answer ‘Yes’ if the roof is sloped and has dormer windows or skylights; otherwise, for a flat roof answer ‘No.’”*

Metrics collected for this case: accuracy, precision, recall, and F1-score—establish the baseline performance of the vision-language model when inferring attic living-space solely from imagery and prompt wording.

Evaluation Process Overview Table 4.1 defines six distinct evaluation conditions obtained by crossing two system-role personas with three zero-shot prompt formulations. Cases 1–3 use the generic “Helpful Assistant” role, while Cases 4–6 employ the more specialized “Expert” persona. Within each persona, prompts P1, P2, and P3 vary in their phrasing and the degree to which they call out specific architectural cues (e.g. slope, dormers, skylights). This factorial design allows us to isolate and quantify the individual and combined effects of persona(system role) framing and prompt wording on the model’s ability to perform the yes/no attic-space classification task. By comparing performance across these six cases, we can determine (a) whether the expert persona consistently outperforms the generic assistant, and (b) which prompt formulation yields the greatest accuracy and recall for habitable-space detection.

Case	System Role	Prompt ID
Case 1	Helpful Assistant	P1
Case 2	Helpful Assistant	P2
Case 3	Helpful Assistant	P3
Case 4	Expert	P1
Case 5	Expert	P2
Case 6	Expert	P3

Tab. 4.1: Evaluation Cases by System Role and Prompt

Evaluation Insights for Dataset 1 To assess how different prompting strategies and system roles influence attic classification from aerial imagery, six prompting cases were evaluated using key performance metrics. Table 4.2 summarizes the accuracy, precision, recall, and F1-scores for each configuration. The corresponding confusion matrices (Figures 4.1a–4.6a) provide additional context for the classification behavior observed.

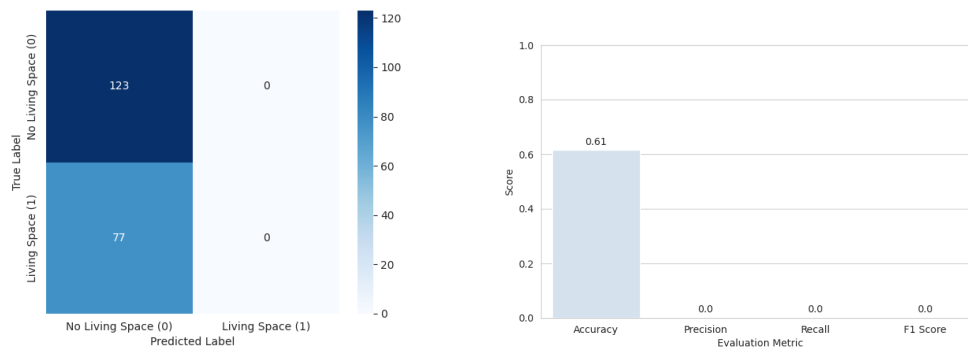
Case-wise Insights:

- **Case 1:** Predicts only "No living space", missing all 77 true positives (4.1a).

Case	System Role	Prompt ID	Accuracy	Precision	Recall	F1-score
1	Helpful Assistant	P1	0.61	0.00	0.00	0.00
2	Helpful Assistant	P2	0.63	1.00	0.04	0.07
3	Helpful Assistant	P3	0.71	0.88	0.29	0.43
4	Expert	P1	0.61	0.00	0.00	0.00
5	Expert	P2	0.74	0.74	0.51	0.60
6	Expert	P3	0.80	0.80	0.64	0.71

Tab. 4.2: Performance metrics across different prompt configurations on Dataset 1.

- **Case 2:** Extremely conservative "Yes" predictions (3 TP, 0 FP), fails to identify most attics (4.2a).
- **Case 3:** Better trade-off—captures 29% of true positives with high precision (4.3a).
- **Case 4:** Expert role alone doesn't help—still predicts only negatives (4.4a).
- **Case 5:** Improved recall (51%) at acceptable precision (0.74) with Expert + Prompt P2 (4.5a).
- **Case 6:** Best overall results—balanced precision and recall yield highest F1 (4.6a).

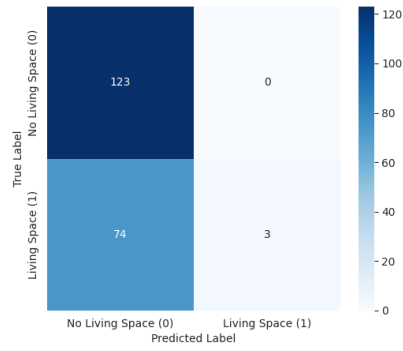


(a) Confusion Matrix for Case 1

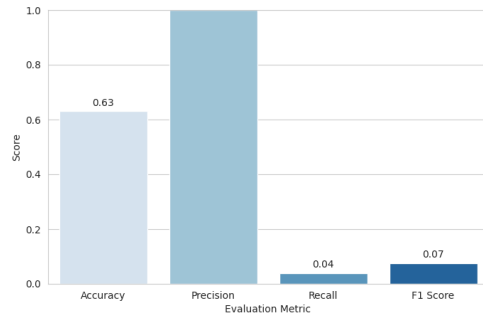
(b) Performance Metrics for Case 1

Fig. 4.1: Case 1 (Helpful Assistant + P1): Confusion matrix and summary of accuracy, precision, recall, and F1.

Comparative Trend of Metrics in Dataset-1 Figure 4.7 plots the progression of four key performance metrics—accuracy, precision, recall, and F1-score—across the six zero-shot inference cases on Dataset 1. Each case combines one of two system roles

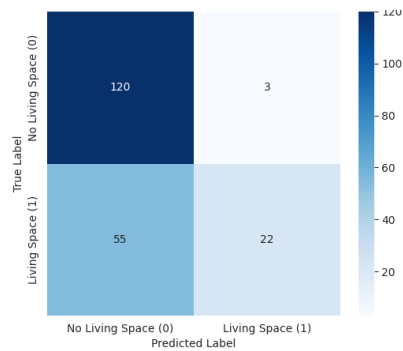


(a) Confusion Matrix for Case 2

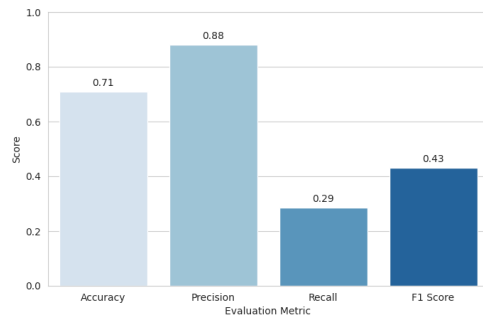


(b) Performance Metrics for Case 2

Fig. 4.2: Case 2 (Helpful Assistant + P2): Confusion matrix and summary of accuracy, precision, recall, and F1.

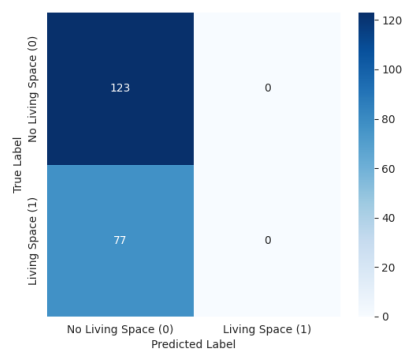


(a) Confusion Matrix for Case 3

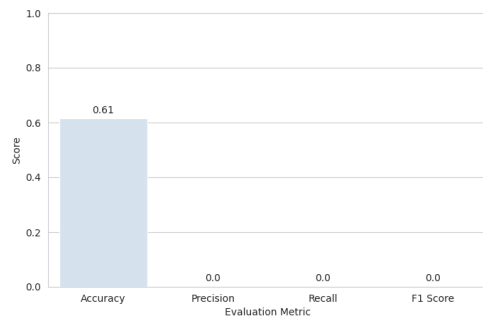


(b) Performance Metrics for Case 3

Fig. 4.3: Case 3 (Helpful Assistant + P3): Confusion matrix and summary of accuracy, precision, recall, and F1.

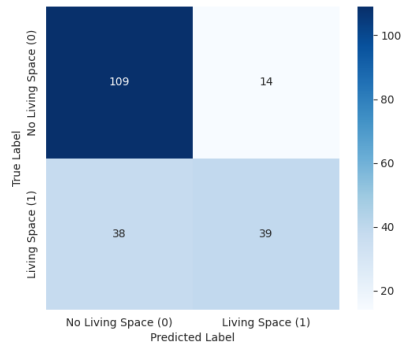


(a) Confusion Matrix for Case 4

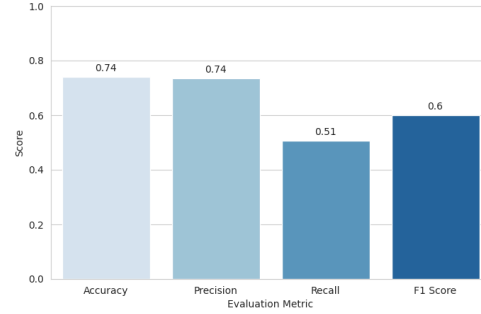


(b) Performance Metrics for Case 4

Fig. 4.4: Case 4 (Expert + P1): Confusion matrix and summary of accuracy, precision, recall, and F1.

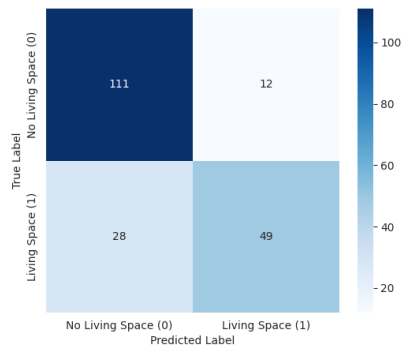


(a) Confusion Matrix for Case 5

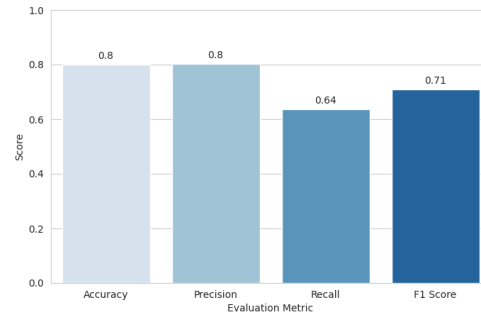


(b) Performance Metrics for Case 5

Fig. 4.5: Case 5 (Expert + P2): Confusion matrix and summary of accuracy, precision, recall, and F1.



(a) Confusion Matrix for Case 6



(b) Performance Metrics for Case 6

Fig. 4.6: Case 6 (Expert + P3): Confusion matrix and summary of accuracy, precision, recall, and F1.

(“Helpful Assistant” or “Expert”) with one of three prompt formulations (P1, P2, or P3).

Case 1 (Helpful Assistant + P1): All metrics collapse to zero “Yes” predictions, yielding 0.61 accuracy (correct “No” classifications) but zero precision, recall, and F1—evidence that the model never flags any image as containing attic space under generic, under-specified prompts.

Case 2 (Helpful Assistant + P2): Precision spikes to 1.00 (three true positives, no false positives) but recall remains near zero (0.04), indicating an overly conservative bias: the model only flags the most obvious examples.

Case 3 (Helpful Assistant + P3): Introducing explicit slope + dormer/skylight cues yields the first trade-off: precision stays high (0.88) while recall climbs to 0.29, lifting F1 to 0.43.

Case 4 (Expert + P1): Even with an expert persona, P1 alone fails to elicit any “Yes” responses, repeating the zero-recall pattern of Case 1.

Case 5 (Expert + P2): The expert framing dramatically improves recall (0.51) at precision 0.74, pushing F1 to 0.60 and accuracy to 0.74—showing that domain-focused persona can unlock latent signal when paired with moderate cues.

Case 6 (Expert + P3): Combining expert persona and the most detailed prompt delivers the best overall performance: accuracy 0.80, precision 0.80, recall 0.64, and F1 0.71.

Taken together, these trends demonstrate that both precise architectural cues in the prompt (P3) and a domain-expert system role are necessary to maximize the model’s ability to detect habitable attic space from aerial images.

Evaluation Insights for Dataset 2 To assess model behavior under Dataset 2, the same set of prompting strategies and personas were applied. Table 4.3 presents the comparative performance across cases.

Case-wise Insights:

- **Case 1:** Predicts only “No living space”, missing all 164 true positives (4.8a).
- **Case 2:** Conservative “Yes” predictions (4 TP, 1 FP); high precision but nearly no recall (4.9a).

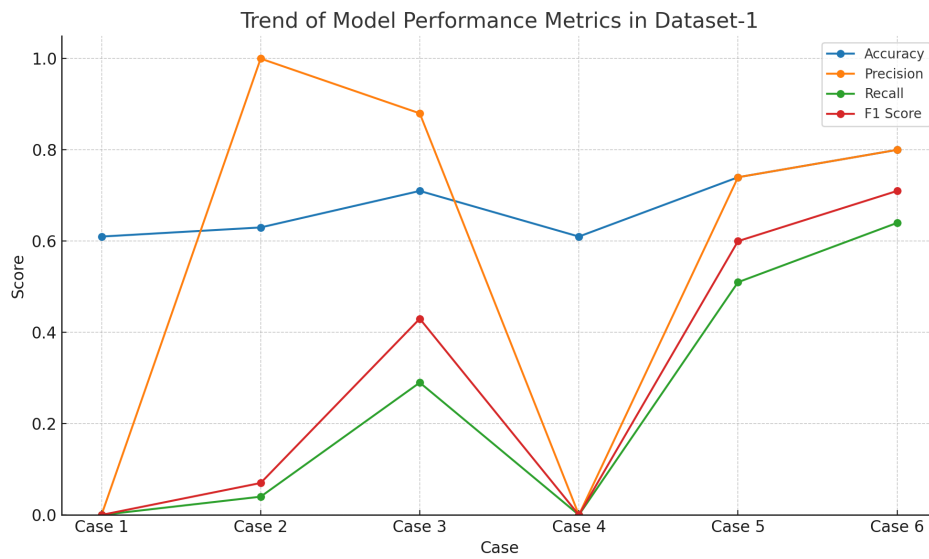


Fig. 4.7: Impact of Prompt Engineering and Persona Framing on Zero-Shot Attic-Space Detection (Dataset 1).

Case	System Role	Prompt ID	Accuracy	Precision	Recall	F1-score
1	Helpful Assistant	P1	0.18	0.00	0.00	0.00
2	Helpful Assistant	P2	0.20	0.80	0.02	0.05
3	Helpful Assistant	P3	0.28	0.75	0.18	0.30
4	Expert	P1	0.18	0.00	0.00	0.00
5	Expert	P2	0.42	0.81	0.37	0.51
6	Expert	P3	0.48	0.79	0.49	0.61

Tab. 4.3: Performance metrics across different prompt configurations on Dataset 2.

- **Case 3:** Addition of slope and dormer cues improves recall to 18%, boosting F1 to 0.30 (4.10a).
- **Case 4:** Expert role alone fails to improve—same as Case 1 (4.11a).
- **Case 5:** Recovers 61 true positives, balancing precision and recall well (4.12a).
- **Case 6:** Best overall—recall near 50%, precision remains high, F1 peaks at 0.61 (4.13a).

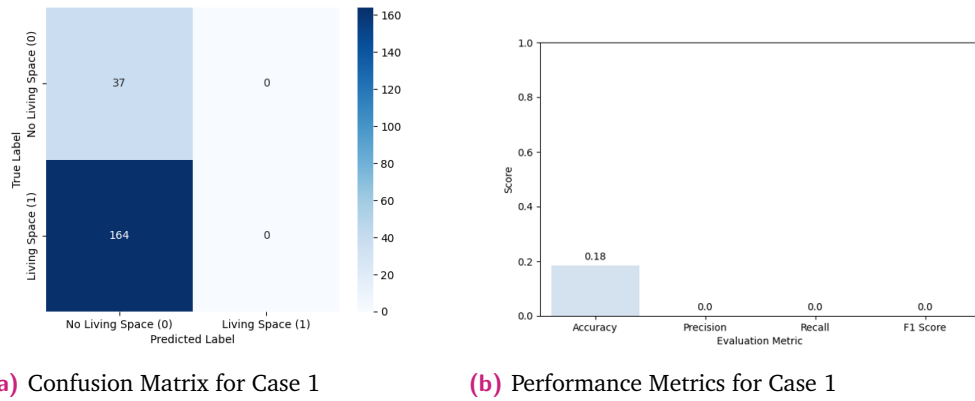


Fig. 4.8: Case 1 (Helpful Assistant + P1): Confusion matrix and summary of accuracy, precision, recall, and F1 for Dataset 2.

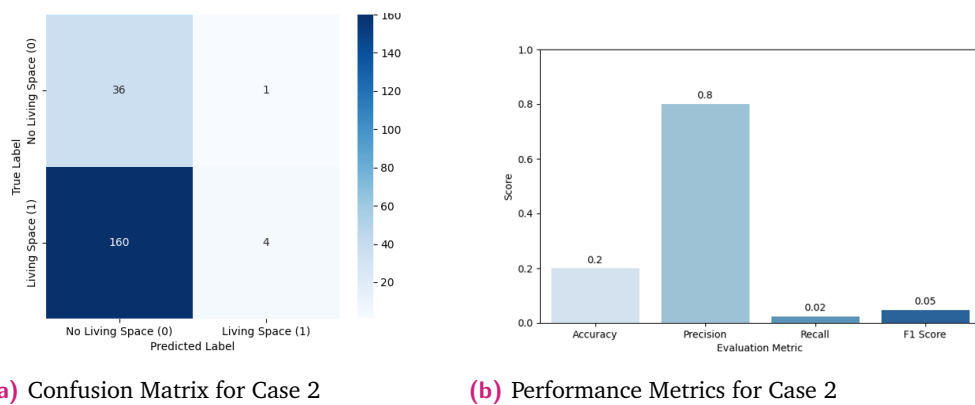
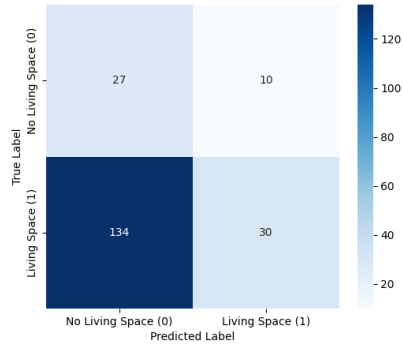
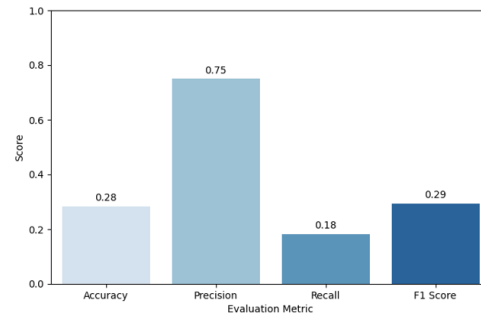


Fig. 4.9: Case 2 (Helpful Assistant + P2): Confusion matrix and summary of accuracy, precision, recall, and F1 for Dataset 2.

Comparative Trend of Metrics in Dataset-2 Figure 4.14 plots the progression of four key performance metrics—accuracy, precision, recall, and F1-score—across the six zero-shot inference cases on Dataset 2. Each case combines one of two system roles (“Helpful Assistant” or “Expert”) with one of three prompt formulations (P1, P2, or P3).

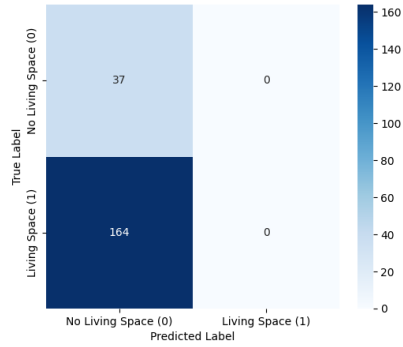


(a) Confusion Matrix for Case 3

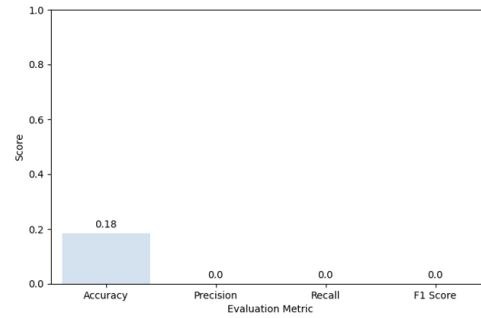


(b) Performance Metrics for Case 3

Fig. 4.10: Case 3 (Helpful Assistant + P3): Confusion matrix and summary of accuracy, precision, recall, and F1 for Dataset 2.

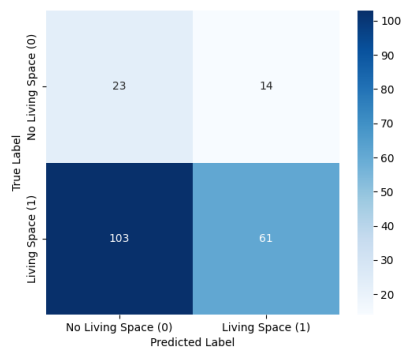


(a) Confusion Matrix for Case 4

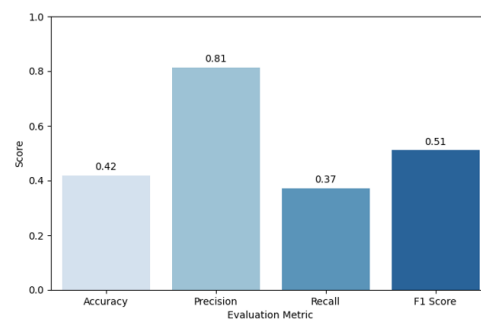


(b) Performance Metrics for Case 4

Fig. 4.11: Case 4 (Expert + P1): Confusion matrix and summary of accuracy, precision, recall, and F1 for Dataset 2.

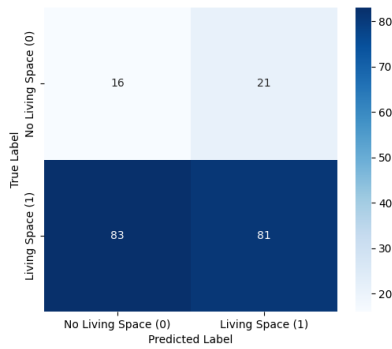


(a) Confusion Matrix for Case 5

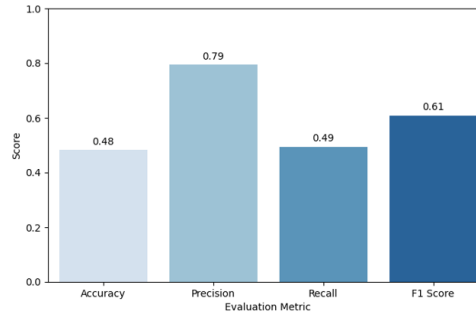


(b) Performance Metrics for Case 5

Fig. 4.12: Case 5 (Expert + P2): Confusion matrix and summary of accuracy, precision, recall, and F1 for Dataset 2.



(a) Confusion Matrix for Case 6



(b) Performance Metrics for Case 6

Fig. 4.13: Case 6 (Expert + P3): Confusion matrix and summary of accuracy, precision, recall, and F1 for Dataset 2.

Case 1 (Helpful Assistant + P1): All metrics collapse to zero “Yes” predictions, yielding 0.18 accuracy (correct “No” classifications) but zero precision, recall, and F1—indicating that the generic prompt P1 fails to elicit any positive detections on purely sloped roofs.

Case 2 (Helpful Assistant + P2): Precision jumps to 0.80 (4 TP, 1 FP) while recall remains negligible (0.02), producing an F1 of 0.05—showing extremely conservative positive labeling.

Case 3 (Helpful Assistant + P3): Incorporating explicit slope and dormer/skylight cues raises recall to 0.18 with precision 0.75, yielding an F1 of 0.30 and modest accuracy improvement to 0.28.

Case 4 (Expert + P1): Even with the expert persona, P1 alone elicits no “Yes” responses, mirroring Case 1’s zero-recall outcome and 0.18 accuracy.

Case 5 (Expert + P2): Expert framing combined with P2 recovers 37

Case 6 (Expert + P3): The most detailed prompt under the expert persona achieves the best balance: accuracy 0.48, precision 0.79, recall 0.49, and F1 0.61, confirming that both precise cues (P3) and an expert role are necessary for optimal attic-space detection on sloped-roof imagery.

Few Shot Prompt Strategy

This evaluation focuses on few-shot prompting applied to Dataset 2 to assess whether the model performs better on a complex dataset, specifically in the scenario where it

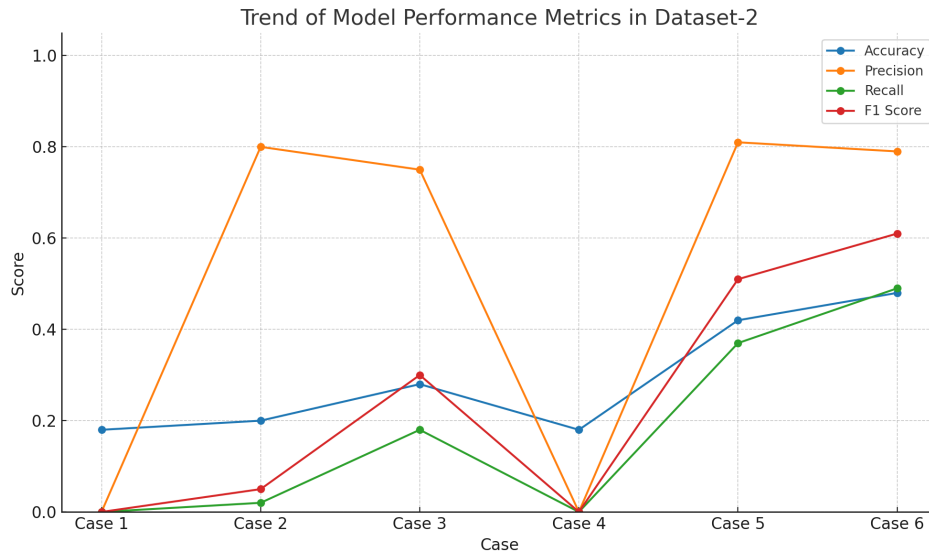


Fig. 4.14: Impact of Prompt Engineering and Persona Framing on Zero-Shot Attic-Space Detection (Dataset 2).

had already achieved the highest accuracy in zero-shot mode, i.e., Case 6. The setup involved providing example input-output pairs, where the input was an aerial image along with the best-performing prompt, P3, and the expected output was a binary “Yes” or “No” answer.

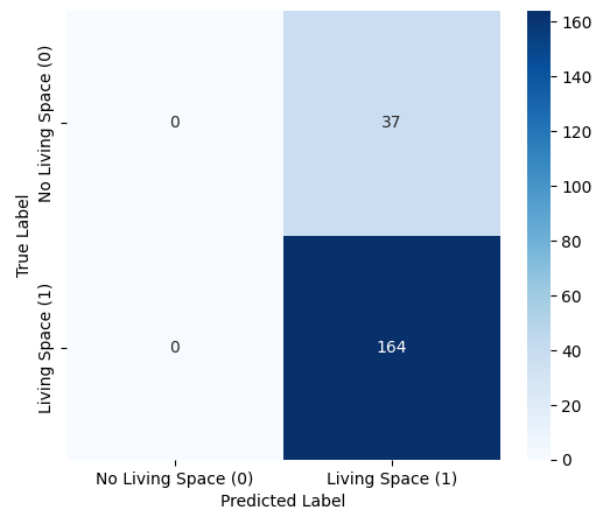


Fig. 4.15: Confusion matrix for few-shot prompting on Dataset 2 using Prompt P3.

The confusion matrix in Figure 4.15 shows that while the model achieved a perfect **recall of 100%**, it also misclassified all negative cases as positive, leading to a lack of true negatives. This behavior indicates a strong tendency to over-predict the presence of attic living space, likely influenced by prompt wording that emphasizes

features such as sloped roofs, dormers, and skylights, architectural traits often associated with habitable areas.

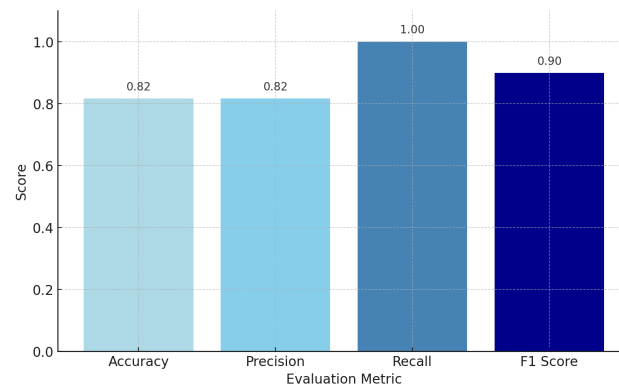


Fig. 4.16: Performance metrics for few-shot prompting using Prompt P3 on Dataset 2.

As illustrated in Figure 4.16, this few-shot setting resulted in an accuracy and precision of 81.6, F1 score of 89.9, and perfect recall of 100. The results demonstrate that the carefully designed prompt P3, when combined with few-shot examples, enabled the model to generalize well in recognizing living space under diverse roof configurations, especially in scenarios where missing positive cases would be more critical than occasionally overestimating.

Despite the trade-off in precision, this case highlights how tailored prompts leveraging domain-specific visual features can significantly enhance few-shot VLM inference in complex aerial imagery.

4.1.2 Experiment 2 – Aerial Roof Feature-Specific: Framing as Binary Classification with Yes/No VQA

A total of 249 aerial images of residential buildings were evaluated in this task. Of these, 166 images have ground-truth “Yes” for attic-living-space usage and 83 have ground-truth “No.” Five key roof attributes were independently queried via binary (Yes/No) prompts:

- **Skylight Prediction:** “Does the roof of the building have any skylights? Answer with only ‘Yes’ or ‘No’ by properly analyzing the image.”
- **Vents/Chimney Prediction:** “Does the roof of the building have any vents or chimneys? Answer with only ‘Yes’ or ‘No’ by properly analyzing the image.”

- **Hipped Roof Type Prediction:** “Is the roof of the building of hipped type? Answer with only ‘Yes’ or ‘No’ by properly analyzing the image.”
- **Gable Roof Type Prediction:** “Is the roof of the building of gable type? Answer with only ‘Yes’ or ‘No’ by properly analyzing the image.”
- **Mixed (Hipped L) Roof Type Prediction:** “Is the roof of the building of mixed (Hipped L) type? Answer with only ‘Yes’ or ‘No’ by properly analyzing the image.”

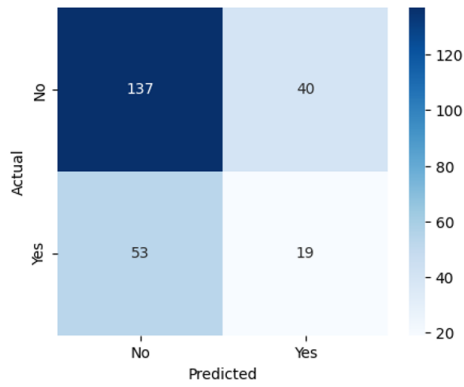
Each of these prompts defines a separate zero-shot binary classification task. Results are summarized via confusion matrices (Figure 4.17) and the associated accuracy, precision, recall, and F1-score metrics (Table 4.4) in the following sections.

Feature	Label	Precision	Recall	F1-Score	Support
Skylights	0	0.72	0.77	0.75	177
	1	0.32	0.26	0.29	72
Vents/Chimneys	0	0.09	0.94	0.16	17
	1	0.98	0.28	0.43	232
Hipped Roof	0	0.79	0.70	0.74	164
	1	0.52	0.65	0.58	85
Gable Roof	0	0.36	0.03	0.05	151
	1	0.38	0.93	0.54	98
Mixed (Hipped L)	0	0.88	0.67	0.76	199
	1	0.33	0.64	0.43	50

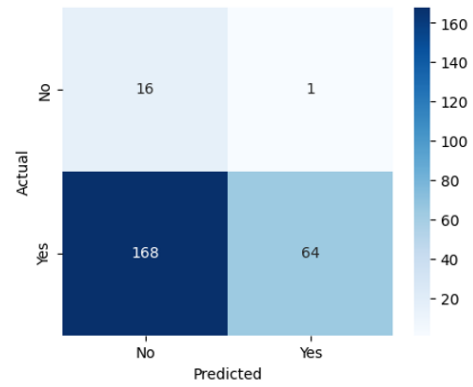
Tab. 4.4: Classification Metrics for Feature-Specific VQA Tasks (Case 2)

The results of Experiment 2 reveal distinct challenges in the model’s ability to classify specific roof features from aerial imagery using binary Yes/No prompts. For skylight detection, the model performed moderately well in identifying negative cases (no skylights), achieving an F1-score of 0.75. However, the model struggled to correctly classify positive instances, with a low F1-score of 0.29. This suggests a tendency to under-predict the presence of skylights, likely due to their small size and variation in appearance.

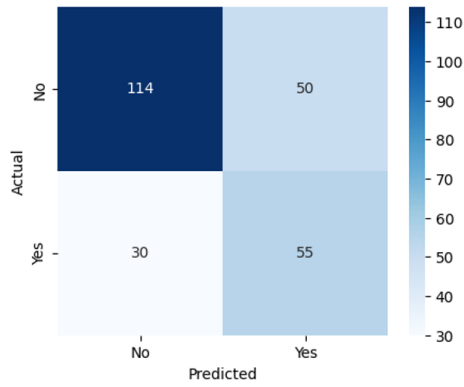
In the case of vents or chimneys, the model exhibited highly imbalanced behavior. It achieved very high precision (0.98) when predicting their presence but with a low recall of 0.28, indicating that many true instances were missed. Conversely, the



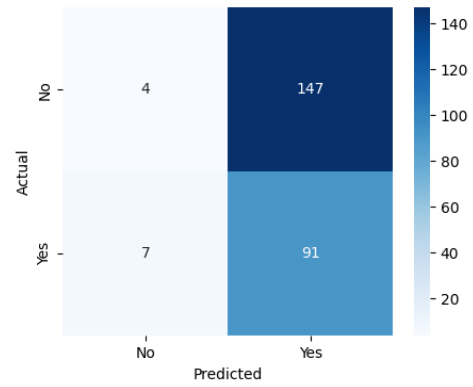
(a) Skylight Detection



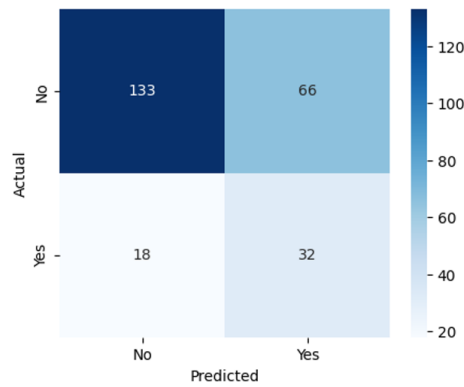
(b) Vents/Chimney Detection



(c) Hipped Roof Type



(d) Gable Roof Type



(e) Mixed (Hipped L) Roof Type

Fig. 4.17: Confusion matrices for feature-specific VQA tasks in Case 2: skylights, vents/chimneys, hipped roof, gable roof, and mixed (Hipped L) roof detection.

model predicted absence with high recall (0.94) but extremely low precision (0.09), showing a bias toward predicting absence unless the feature was strongly evident.

For the hipped roof category, the model displayed more balanced performance, with F1-scores of 0.74 and 0.58 for the negative and positive classes, respectively. This indicates that it could reasonably distinguish hipped roofs from other types based on aerial patterns.

In contrast, predictions for gable roofs showed strong imbalance. The model achieved high recall (0.93) and moderate F1-score (0.54) for the positive class, but almost completely failed to identify negative examples, yielding a recall of only 0.03 and F1-score of 0.05. This reveals a strong tendency to over-predict gable roofs, possibly due to their more frequent or visually prominent structure in the dataset.

Finally, for the mixed (Hipped L) roof type, the model performed moderately, with an F1-score of 0.76 for negative and 0.43 for positive predictions. While it could identify the absence of such roofs with reasonable accuracy, it showed reduced precision in detecting their actual presence—likely reflecting the complexity of distinguishing such composite shapes.

Overall, the experiment demonstrates that the model handles prominent roof geometries more reliably than smaller, less visually consistent elements. Class imbalance and visual ambiguity were key factors affecting the quality of prediction across different features.

4.2 Prompt Based Inference Utilizing Only Tabular Data

In this experiment, imagery is set aside and attic space usage is predicted solely from tabular roof attributes for a group of 745 buildings. The distribution of the dataset is explained in 3.3 The analysis primarily focused on the distribution of buildings based on `roof_height_max` and `slope_max`, as described in the methodology. These two attributes emerged as key indicators for distinguishing buildings with potential attic space usage.

4.2.1 Prompt Formulations

Two templated text prompts were used to query the model's tabular-data inference:

- **Prompt (AND):** “Given the following building attributes:

- Building ID: {row['gid']}
- Maximum roof slope: {row['slope_max']} degrees
- Maximum roof height: {row['roof_height_max']} meters

If the maximum roof slope is greater than or equal to 30 degrees **AND** the maximum roof height is greater than or equal to 4 meters, predict that the attic is used as a living space.

At the beginning of your response, output only ‘Yes’ or ‘No’ on a new line. Do not repeat the input. Do not explain.”

- **Prompt (OR):** “Given the following building attributes:

- Building ID: {row['gid']}
- Maximum roof slope: {row['slope_max']} degrees
- Maximum roof height: {row['roof_height_max']} meters

If the maximum roof slope is greater than or equal to 30 degrees **OR** the maximum roof height is greater than or equal to 4 meters, predict that the attic is used as a living space.

At the beginning of your response, output only ‘Yes’ or ‘No’ on a new line. Do not repeat the input. Do not explain.”

4.2.2 Evaluation by Roof Type and Tabular Prompts

Each prompt is applied to three subsets of the data—gable roofs, hipped roofs, and the full dataset—resulting in six distinct model runs:

1. Gable roofs with the AND prompt
2. Gable roofs with the OR prompt
3. Hipped roofs with the AND prompt
4. Hipped roofs with the OR prompt
5. All roofs with the AND prompt
6. All roofs with the OR prompt

Each configuration yields a corresponding confusion matrix and standard evaluation metrics, including accuracy, precision, recall, and F1-score, as illustrated in Figures 4.18.

The following sections present a detailed comparison of the accuracy, precision, recall, and F1-score obtained across the two prompt types (AND / OR) for each of the three data subsets: gable roofs, hipped roofs, and the full dataset.

Subset	Case	Accuracy	Precision	Recall	F1-Score
Gable Roofs	AND	0.74	1.00	0.48	0.65
	OR	0.83	1.00	0.77	0.87
Hipped Roofs	AND	0.75	0.99	0.45	0.62
	OR	0.80	0.99	0.63	0.77
All Roofs (Combined)	AND	0.74	0.99	0.47	0.64
	OR	0.82	0.99	0.71	0.83

Tab. 4.5: Tabular-Only Inference Metrics by Roof Subset and Case

4.2.3 Key Findings

The evaluation demonstrates that the logical structure of the prompt (AND vs. OR) significantly affects model performance across all roof subsets. Prompts using the OR condition consistently outperformed their AND counterparts in terms of recall and F1-score, indicating that the more inclusive logic better captured potential attic usability.

For **gable roofs**, the OR prompt yielded a notably higher F1-score (0.87) compared to the AND prompt (0.65), with a substantial gain in recall from 0.48 to 0.77, while maintaining perfect precision (1.00) in both cases. This suggests that gabled structures benefit from the relaxed conversion criteria, allowing the model to identify more true positives without sacrificing correctness.

A similar pattern was observed for **hipped roofs**, where the F1-score improved from 0.62 (AND) to 0.77 (OR), with recall increasing from 0.45 to 0.63. Precision remained high (0.99) in both scenarios, indicating that while the model remained cautious, the OR logic enabled broader detection of convertible attics.

When evaluating **all roofs combined**, the OR prompt again led to better overall performance, achieving an F1-score of 0.83 versus 0.64 under the AND condition. The improvement in recall (from 0.47 to 0.71) confirms that relaxing the constraints allowed the model to generalize better across varied roof types.

Overall, these results indicate that the OR prompt formulation—capturing either slope or height thresholds or the presence of dormers—is more effective for identifying buildings with existing or potential attic living space when only tabular metadata is used.

4.3 Prompt Based Inference Combining Tabular Data and Imagery

This section presents the evaluation of prompt-based inference that integrates both tabular data and aerial imagery, aiming to gain insights about the prediction of attic usability through multimodal reasoning. It is performed only for the combined case mentioned in 4.2. By combining visual cues, such as roof geometry and feature presence, with structured roof metadata, the model is prompted for tabular data and images and the final prediction is combined using the outputs. The evaluation compares performance against tabular-only and image-only baselines, highlighting whether the inclusion of visual context provides measurable improvements in classification accuracy and feature recognition.

4.3.1 Prompt Formulations

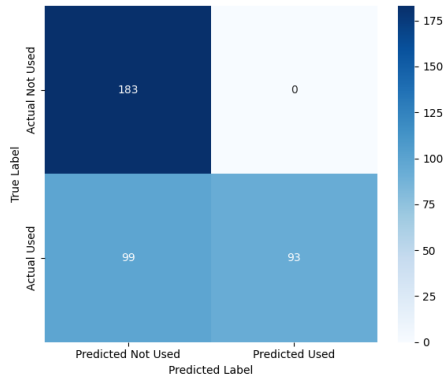
Following prompts were used for the actual attic usage case :

- **Tabular Prompt (AND):** *“Given the following building attributes:
– Building ID: {row['gid']}
– Maximum roof slope: {row['slope_max']} degrees
– Maximum roof height: {row['roof_height_max']} meters

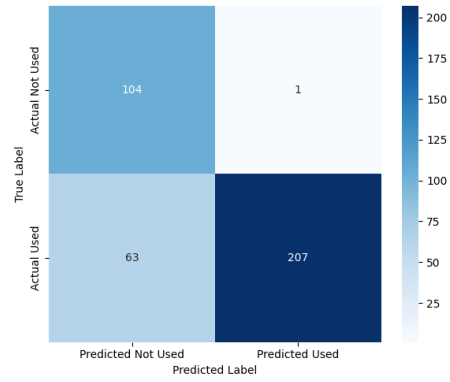
If the maximum roof slope is greater than or equal to 30° **AND** the maximum roof height is greater than or equal to 4 m, predict that the attic is used as a living space.

At the beginning of your response, output only ‘Yes’ or ‘No’ on a new line. Do not repeat the input. Do not explain.”*
- **Visual Dormer-Detection Prompt:** *“Does the building roof have dormers? Answer with a yes or no.”*

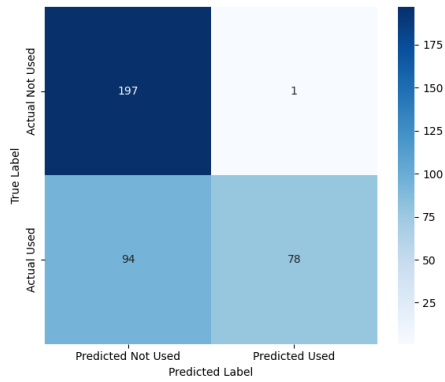
Following prompts were used for potential conversion case:



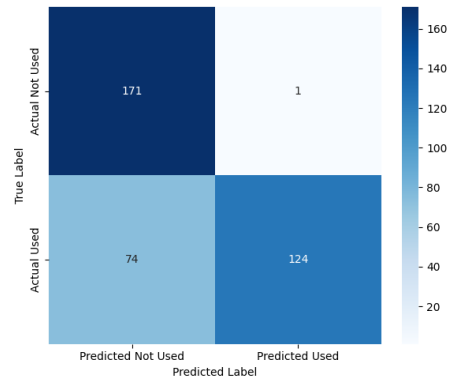
(a) Gable Roofs – AND Rule



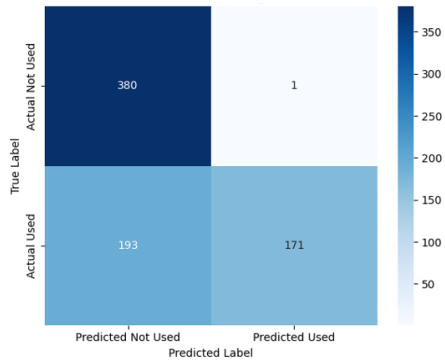
(b) Gable Roofs – OR Rule



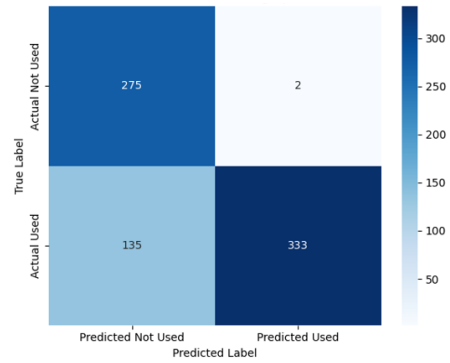
(c) Hipped Roofs – AND Rule



(d) Hipped Roofs – OR Rule



(e) All Roofs – AND Rule



(f) All Roofs – OR Rule

Fig. 4.18: Confusion matrices for tabular-data attic-space inference under AND vs. OR prompts, across gable, hipped, and combined roof subsets.

- **Tabular Prompt (OR):** *“Given the following building attributes:*
 - *Building ID: {row['gid']}*”
 - *Maximum roof slope: {row['slope_max']}* degrees
 - *Maximum roof height: {row['roof_height_max']}* meters

If the maximum roof slope is greater than or equal to 30° OR the maximum roof height is greater than or equal to 4 meters, predict that the attic is used as a living space.

At the beginning of your response, output only ‘Yes’ or ‘No’ on a new line. Do not repeat the input. Do not explain.”
- **Visual Dormer-Detection Prompt:** *“Does the building roof have dormers? Answer with a yes or no.”*

4.3.2 Evaluation Results

Figure 4.19a shows the confusion matrix under the AND-based ground truth. Figure 4.19b displays the confusion matrix under the OR-based ground truth. Table 4.6 summarizes the performance of the combined inference approach, which leverages both tabular data and aerial imagery to predict attic usability. Compared to the tabular-only case, the integration of visual cues led to more balanced results across evaluation metrics. For the AND prompt condition, the model achieved an accuracy of 0.69 and an F1-score of 0.68, indicating steady but conservative predictions. Under the OR condition—which allows more flexibility in identifying convertible attics—the F1-score improved slightly to 0.71, driven by a higher precision of 0.79. These results suggest that while the inclusion of imagery offers marginal gains in recall and overall balance, the improvement is less pronounced than expected, likely due to challenges in effectively aligning visual and structured data through prompt-based reasoning. This might be due to samples where the insights from image i.e., there might be a dormer present, but the corresponding tabular thresholds don't satisfy. In such case the model is in ambiguity to make prediction and leads to decrease in accuracy in comparison to the tabular only inference.

Together, these results illustrate that integrating visual dormer detection with simple tabular thresholds provides a tunable inference strategy:

- *Strict (AND)* mode guarantees reliability when both data modalities align,
- *Potential (OR)* mode maximizes coverage for identifying candidate conversion sites.

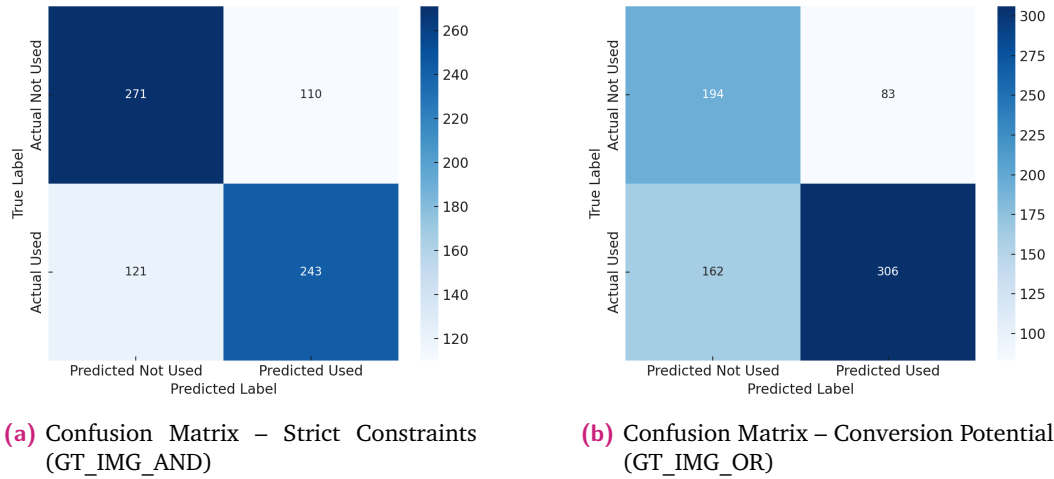


Fig. 4.19: Confusion matrices for tabular and imagery inference under the AND-based (Case 1) and OR-based (Case 2) ground-truth definitions.

Subset	Case	Accuracy	Precision	Recall	F1-Score
All Roofs (Combined)	AND	0.69	0.69	0.67	0.68
	OR	0.68	0.79	0.65	0.71

Tab. 4.6: Combined Inference Metrics for the entire dataset

Discussion

This chapter presents a critical interpretation of the experimental results obtained in the course of evaluating vision-language models (VLMs) for the multimodal analysis of physical structures using aerial imagery and tabular metadata. The goal is to contextualize the findings with respect to the research questions outlined in earlier chapters, and to explore the implications of the proposed approach for real-world applications such as urban planning, energy assessment, and geospatial analytics.

The discussion is organized around the three central research questions that guided the study. First, it examines the ability of open-source VLMs to accurately extract fine-grained architectural features such as dormers and skylights from aerial images of residential rooftops. Second, it explores the integration of visual and tabular data using prompt-based methods inspired by Visual Question Answering (VQA), and assesses whether multimodal inference offers measurable advantages over single-modality approaches. Finally, it evaluates the reliability of the model outputs by comparing them against predefined ground truth conditions and analyzing the results using standard classification metrics.

5.1 Interpreting Visual Feature Extraction from Aerial Imagery

This section discusses the ability of vision-language models (VLMs) to detect and interpret architectural features from aerial imagery, as evaluated in the context of residential buildings in Kassel. The focus lies on analyzing the model's performance in identifying dormers and skylights two key indicators used in classifying attic usability. The results are further interpreted across different roof types (gable and hipped), and the influence of prompt design and conditioning strategies is examined. Finally, the limitations observed in visual-only inference are critically assessed.

5.1.1 Accuracy of Feature Detection

The evaluation shows that VLMs can reasonably detect prominent roof features, particularly dormers, with moderate precision. For image-only prompts, detection performance varies across features and roof geometries. Dormers were more consistently identified than skylights, likely due to their larger size and more distinguishable structural outlines in aerial imagery. Standard classification metrics such as accuracy, precision, recall, and F1-score were used to quantify the performance.

The best results for image-only inference were observed under relaxed prompting conditions, where descriptive cues led to higher recall. However, these gains occasionally came at the cost of precision, indicating a trade-off between identifying all relevant instances and avoiding false positives. This balance is especially critical in downstream tasks such as attic usability classification, where overestimation may lead to misleading assessments.

5.1.2 Influence of Roof Type on Model Performance

Model performance varied notably across roof types. In general, hipped roofs yielded more accurate predictions than gable roofs across all evaluated features. This can be attributed to their more uniform geometry and symmetrical layout, which likely provided more consistent visual cues for the model to learn from. In contrast, gable roofs often exhibit sharper contours and more varied roof extensions, such as dormers and skylights, which introduce visual complexity and increased variation in appearance.

5.1.3 Effect of Prompt Design and Role Conditioning

Prompt formulation had a significant impact on the model's interpretive behavior. System-role prompts that guided the model to “analyze the image carefully for roof structures such as dormers or skylights” tended to yield more focused responses compared to generic image caption prompts. Additionally, the inclusion of positional cues (e.g., “on the front side of the roof”) improved the model's attention to relevant regions of the image. Few-shot prompting, where example image-prompt-response pairs were provided during inference, further improved performance relative to the zero-shot baseline. This enhancement was particularly evident in tasks involving complex or ambiguous roof features, indicating that the model benefits from explicit examples to ground its reasoning.

Chain-of-thought prompting was not explicitly used in this research, but evidence suggests that introducing stepwise reasoning (e.g., “First detect the roof shape. Then determine if any structures protrude.”) may further improve interpretability and robustness. While prompting did not eliminate hallucinations or ambiguity, it played a critical role in guiding the model toward more relevant visual features.

5.1.4 Limitations of Vision-Language Models in Visual Inference

Despite promising results, several limitations emerged in the use of VLMs for aerial image interpretation. First, visual ambiguity due to occlusions, shadows, or low-resolution imagery led to frequent misclassifications, particularly for skylights. Second, the lack of domain-specific pretraining in open-source models made them prone to hallucinating features when visual signals were weak or uncertain. In some cases, the model inferred the presence of dormers even when none existed, likely due to over-reliance on pattern repetition learned from generic datasets.

Another limitation is the absence of spatial grounding. The model does not inherently localize features spatially within the image, making it difficult to confirm whether the described feature corresponds to the actual location on the roof. This weakens the interpretability of outputs in applications requiring geometric precision.

Overall, while VLMs show the potential to extract informative features from aerial imagery, their capabilities remain constrained by data quality, domain specificity, and the need for structured visual grounding.

5.2 Linking Aerial Imagery with Tabular Metadata

This section addresses how tabular data was integrated with aerial imagery using prompt-based multimodal techniques inspired by Visual Question Answering (VQA). The objective was to determine whether the inclusion of metadata such as slope, height, and dormer presence could improve the inference of structural characteristics compared to relying on imagery alone.

5.2.1 Methods for Image - Table Association

To enable multimodal reasoning, each image was associated with a corresponding row in the tabular dataset using unique image identifiers. These associations allowed

prompts to explicitly reference both the visual content and relevant numerical or categorical attributes (e.g., “The slope is 35 degrees, and the height is 4.5 meters”). By embedding structured data into the prompt context, the model was given access to additional information that could resolve visual ambiguity or enhance reasoning.

5.2.2 Effectiveness of Prompt-Based Multimodal Inference

The integration of tabular attributes into the prompting process generally improved the quality of the model’s responses. In particular, the use of OR-based logical constraints in prompts (e.g., “slope ≥ 30 OR height ≥ 4 OR dormer = 1”) led to higher recall without significantly reducing precision. This suggests that multimodal prompting allows the model to infer structure usability more robustly, especially in borderline cases where the image alone is inconclusive. The model was also better at justifying its output when prompted with both modalities, offering more contextually grounded responses.

5.2.3 Comparison: Multimodal vs Single-Modality Performance

The comparative evaluation between single-modality (tabular-only) and multimodal (tabular + image) inference reveals that the multimodal approach did not consistently outperform its tabular-only counterpart. This performance gap can be attributed to inconsistencies between visual cues extracted from aerial images and the rule-based thresholds defined in the tabular ground truth.

In many instances, the image-based insights appeared misaligned with structured metadata such as roof slope or height. The vision-language model might identify roof features like dormers or skylights that visually suggest livable attic space, but these features may not satisfy the predefined numerical constraints in the tabular ground truth. As a result, such cases were penalized during evaluation, leading to lower measured performance. This finding underscores the challenge of integrating vision and structured reasoning without tightly coupled alignment between modalities, particularly when rule-based tabular logic dominates the ground truth definition.

5.2.4 Challenges in Multimodal Alignment and Reasoning

Despite performance improvements, several challenges were observed. One key issue was prompt sensitivity: small changes in prompt phrasing sometimes led to

inconsistent outputs. Additionally, hallucinations persisted when visual and tabular data provided conflicting signals. In such cases, the model occasionally disregarded metadata in favor of visual patterns, highlighting the need for improved alignment mechanisms. Finally, the model lacked a built-in grounding mechanism to verify that specific features mentioned in the prompt were visually present, limiting the trustworthiness of multimodal justifications.

5.3 Evaluation and Validation Against Ground Truth

This section examines how the outputs of the inference system were evaluated against predefined ground truth conditions. It focuses on metric selection, performance variation across different roof types and features, and the comparative analysis of single-modality and multimodal inference.

5.3.1 Selection of Evaluation Metrics

To ensure consistent and interpretable assessment, standard metrics including accuracy, precision, recall, and F1-score were used across all experiments. These metrics were computed for both binary classification tasks (e.g., attic usable or not) and feature-specific evaluations (e.g., dormer presence). Confusion matrices provided additional insights into the nature of errors made by the model under different inference modes.

5.3.2 Performance Across Logical Constraints (AND vs OR)

Two types of logical constraints were used to define the ground truth: strict (AND) and relaxed (OR). The OR-based conditions led to higher recall, reflecting a more permissive classification strategy that captures marginal cases. In contrast, AND-based inference yielded higher precision but suffered from lower recall, especially when the visual cues were ambiguous. These results emphasize the importance of selecting appropriate logical conditions depending on the desired application risk-sensitive systems may favor precision, while exploratory tasks may prioritize recall.

5.3.3 Differences by Roof Type and Feature Class

Performance metrics revealed that gable roofs consistently achieved better classification outcomes than hipped roofs. This was true across both modalities and logical constraints. Dormers were more reliably detected than skylights, likely due to their size and shape prominence. In the tabular context, attributes like slope and height were used effectively for classification, though their interpretability was limited in isolation.

5.3.4 Observed Strengths and Failure Cases

The system showed particular strength in identifying structures that matched typical patterns seen in training data. However, it struggled with edge cases such as small dormers, atypical skylights, or roofs partially covered by vegetation. Another limitation was the model's occasional tendency to hallucinate features when visual evidence was ambiguous. These failure modes point to the need for incorporating uncertainty estimation or confidence thresholds in future versions of the system.

5.4 Deviation from Initial Implementation Plan

While the initial thesis proposal included the development of an interactive chatbot interface for querying building information using multimodal prompts, the focus of the research evolved over time. As the use case shifted toward assessing model performance and understanding the limitations of vision-language models in structured inference tasks, the emphasis moved toward systematic evaluation and benchmarking rather than full-stack application development. Additionally, although the proposal intended to utilize open-source datasets, the final implementation relied on proprietary building-level data due to the lack of publicly available datasets with comparable spatial, semantic, and structural detail. Preparing this dataset for multimodal inference proved to be a complex and time-consuming task, involving significant preprocessing, cleaning, and alignment between imagery and tabular metadata. Furthermore, only a limited number of publicly accessible datasets exhibit similar characteristics, limiting the potential for external validation or cross-dataset benchmarking. These factors influenced both the research direction and the depth of evaluation presented in this thesis.

5.5 Future Work

Building on the findings of this thesis, several avenues for future research and development are being considered to enhance the performance and applicability of multimodal AI systems for building analysis from aerial imagery.

First, the use of higher-resolution aerial imagery is expected to improve model accuracy, particularly in detecting fine-grained architectural features like skylights, dormers, chimneys, or roof textures. Future studies can incorporate commercial-grade or drone-captured imagery, which offers finer spatial granularity than public satellite datasets. With better visual detail, Vision-Language Models (VLMs) are likely to deliver more reliable and interpretable predictions.

Second, the integration of 3D building models is envisioned to play a crucial role in enhancing spatial reasoning. Techniques such as photogrammetry or LiDAR can be used to generate Level of Detail (LoD)-compliant 3D representations, which provide rich geometric and semantic context. These models will allow for a volumetric understanding of the structure, enabling more accurate inference about internal space usage, roof pitch calculations, and overall building morphology. Future implementations may combine CityGML-based 3D data with language models to create spatio-semantic prompts.

Third, fine-tuning of VLMs on domain-specific datasets is considered an important next step. While this study relied on zero-shot inference, future work can involve curating labeled datasets of roof features and urban structures for supervised fine-tuning. This adaptation will help overcome the current generalization limitations observed in open-source VLMs and significantly boost their domain alignment and reliability.

Conclusion

This thesis explored the potential of Generative AI, particularly Vision-Language Models (VLMs), to analyze aerial imagery of physical structures and link this visual data with corresponding tabular attributes. The work was motivated by the growing need to automate and scale energy diagnostics, urban infrastructure assessments, and remote sensing analysis using multimodal AI systems.

The central problem addressed was whether and how modern VLMs, when combined with prompt engineering, could infer fine-grained architectural features (such as dormers and skylights) from aerial roof images and contribute to tasks like attic usability classification. To this end, a comprehensive evaluation framework was developed. It included the shortlisting of promising open-source VLMs through leaderboard analysis, benchmarking, and architectural review. Models were then subjected to zero-shot tasks involving image-only and image-tabular hybrid prompts, with a focus on interpretable outputs.

Key findings from the study show that VLMs such as Qwen2.5-VL-7B-Instruct are capable of recognizing complex roof features when guided by well-structured prompts. The study also demonstrated that integrating visual cues with structured data, such as roof slope, type, and height, significantly enhances the interpretability and accuracy of classification tasks. Additionally, prompt formulation played a crucial role in model performance, revealing that linguistic clarity and the presence of system-role instructions can influence outcomes.

Despite these encouraging results, the work is not without limitations. The image dataset, though carefully curated, was constrained by resolution and clarity, limiting the model's ability to detect subtle architectural features. Further, the lack of domain-specific fine-tuning for VLMs meant that outputs occasionally lacked consistency or spatial grounding. Computational constraints also limited the extent of large-scale experimentation.

Nevertheless, this thesis makes several contributions: (i) it provides a methodology for evaluating and applying VLMs in remote sensing contexts, (ii) it showcases the power of prompt engineering in enhancing zero-shot visual inference, along with

few shot prompting (iii) it offers a proof-of-concept for a methodology that can link aerial imagery with tabular data in an interactive setting.

In conclusion, this work demonstrates the viability of combining Generative AI techniques with geospatial and structured data to analyze physical structures in urban environments. It lays the foundation for more scalable, intelligent systems for energy diagnostics and urban analysis, while also highlighting several important directions for future work, including model fine-tuning, automated prompt optimization, and expansion into 3D spatial representations.

The full implementation and experimental notebooks are available at the following GitHub repository: [Jen25]

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